

Lectures on Hawking Radiation

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Abstract

Hawking radiation is sometimes described as an effect of quantum gravity. It is better understood as a consequence of quantum field theory on a fixed background, and ultimately of a single geometrical fact: outgoing rays near a forming horizon are redshifted exponentially. These lectures are organised around that observation. After the classical laws of black hole mechanics and a short treatment of the Unruh effect, we derive the Hawking temperature step by step, then compare the Euclidean, anomaly and tunnelling derivations to see which ingredients each of them actually requires. The numbers matter as much as the formalism. A black hole of one solar mass radiates at about 6×10^{-8} K and would need some 10^{67} years to evaporate, whereas a primordial black hole of 5×10^{14} g would be completing its evaporation today. The longest lecture concerns the information problem, from Hawking's 1976 argument to the Page curve and the island calculations that reproduce it. The course closes with what can be tested: analogue horizons in Bose–Einstein condensates, and the searches for evaporating black holes in astrophysical data and at colliders. Each lecture ends with problems and with annotated suggestions for further reading.

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Lecture 1 The temperature of a black hole

In 1972 Jacob Bekenstein, then a graduate student at Princeton, proposed that a black hole carries an entropy proportional to the area of its horizon (Bekenstein, 1972, 1973).¹ The objection was immediate, and its most careful statement came from Bardeen et al. (1973). A body with entropy S and energy M has a temperature, $T^{-1} = \partial S / \partial M$. A body with a temperature radiates. A black hole in classical general relativity absorbs everything and emits nothing, so its temperature is zero, and its area cannot be a thermodynamic entropy. Within two years Hawking (1974, 1975) had shown that the objection fails, and that what defeats it is quantum mechanics. These lectures are about that result and about the problems it has raised in the half century since.

1.1 A puzzle from black hole thermodynamics

Classically, a black hole is a perfect absorber, its horizon a one-way membrane. Yet by 1973 it was clear that black holes obey laws of the same form as the laws of thermodynamics (Bardeen et al., 1973). The surface gravity κ of a stationary black hole is constant over its horizon, as the temperature of a body in equilibrium is uniform. The area A of the horizon never decreases, as the entropy of an isolated system never decreases. Small changes in the mass, area, angular momentum and charge are related by an identity with the form of the first law, in which $\kappa/8\pi$ stands where the temperature should be and A where the entropy should be. The four laws are stated precisely in Lecture 2.

Bekenstein took the analogy literally. Bardeen, Carter and Hawking regarded $\kappa/8\pi$ and A as analogues of temperature and entropy, distinct from them. Within classical general relativity they were right: either the analogy is formal, or the theory is missing an ingredient.

The missing ingredient is quantum mechanics. When quantum fields are placed on the geometry of a black hole formed by collapse, the black hole emits thermal radiation at the

¹The question came from his adviser, John Wheeler, who put it as a joke about a crime: mixing a cup of hot tea with a cup of cold tea increases the entropy of the world, and dropping both into a black hole would seem to hide the evidence. Bekenstein’s answer was that the black hole itself must carry the entropy.

temperature

$$T_H = \frac{\hbar \kappa}{2\pi c k_B} \xrightarrow{\text{Schwarzschild}} \frac{\hbar c^3}{8\pi G M k_B} \simeq 6.17 \times 10^{-8} \text{ K} \left(\frac{M_\odot}{M} \right). \quad (1.1)$$

Hawking had expected to find something quite different.² Once the temperature is known, the first law fixes the constant that Bekenstein could not, and the entropy is the Bekenstein–Hawking entropy,

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{k_B A}{4\ell_P^2}, \quad \ell_P = \left(\frac{G\hbar}{c^3} \right)^{1/2}. \quad (1.2)$$

Both numbers are extreme. A black hole of one solar mass has a temperature of 6.2×10^{-8} K, nearly eight orders of magnitude below that of the cosmic microwave background, and an entropy of $1.0 \times 10^{77} k_B$.

1.2 A first estimate

Much of Eq. (1.1) can be guessed before any calculation. A Schwarzschild black hole has a single parameter, its mass M . From M , Newton’s constant and the speed of light one can form a single time scale, the light-crossing time of the horizon, $t_M = GM/c^3$, which is $4.9 \mu\text{s}$ for a solar mass. Classical physics provides no way to convert a time into a temperature. Quantum mechanics does, since $\hbar/t_M$ is an energy, and the only temperature that can be built from the available quantities is therefore

$$k_B T \sim \frac{\hbar}{t_M} = \frac{\hbar c^3}{GM}. \quad (1.3)$$

For a solar mass this gives 1.6×10^{-6} K. The factor $1/8\pi$ in Eq. (1.1), which only the full calculation can supply, brings it down to 6.2×10^{-8} K.

The estimate already contains two facts that survive the full calculation. The temperature vanishes in the classical limit $\hbar \rightarrow 0$, which is why classical general relativity could not have found it. It is also inversely proportional to the mass: small black holes are hot and large ones cold, so that a radiating black hole heats up and radiates faster, a runaway examined in Lecture 5. The same reasoning applied to the entropy, which is dimensionless in units of k_B and must be built from the area and the fundamental constants, gives $S \sim A/\ell_P^2$, in agreement with Eq. (1.2); the factor $1/4$ again requires the calculation.

1.3 A heuristic picture and its limitations

The usual picture of the origin of the radiation goes back to Hawking himself. The quantum vacuum contains fluctuations that may be described as virtual particle–antiparticle pairs, which in flat spacetime annihilate within the time allowed by the energy–time uncertainty relation. Near a horizon, one member of a pair may cross the horizon before annihilation takes place, while the other escapes to infinity as a real particle. Energy conservation then requires the member that falls in to carry negative energy as measured from infinity,³ and the black hole loses mass. The picture even reproduces the scale of Eq. (1.3): a pair can be separated by the horizon only if it survives for a time of order t_M , which requires an energy of order $\hbar/t_M$.

²He was led to the problem partly by Zel’dovich and Starobinsky, who argued during his visit to Moscow in 1973 that a rotating black hole should emit in its superradiant modes (Section 2.5). Improving their calculation, he found to his surprise that a non-rotating black hole emits as well.

³This is possible because the Killing vector ∂_t , which defines energy at infinity, is spacelike inside the horizon. The Penrose process of Section 2.2 exploits the same freedom in the ergoregion of a Kerr black hole.

It should not be taken further than that. A quantum of energy $\hbar\omega = x k_B T_H$ has wavelength

$$\lambda = \frac{2\pi c}{\omega} = \frac{8\pi^2}{x} r_s, \quad r_s = \frac{2GM}{c^2},$$

so that at the peak of the energy spectrum of a black body, $x \approx 2.8$, the wavelength is about 28 Schwarzschild radii. A particle of this wavelength cannot be localised in a thin layer just outside the horizon, which is what the picture suggests (Remark 5.1). Nor does the picture explain why the spectrum is thermal. The correct statement, which we develop in Lecture 3, is that the state of a quantum field that is regular across a forming horizon necessarily contains correlations between modes inside and outside the horizon, and that the outside modes, considered on their own, are thermally populated. The pair picture is best regarded as a mnemonic for this entanglement. We shall not use it for anything else.

1.4 How the course is organised

Lectures 2 and 3 are the core of the course: the classical laws of black hole mechanics, the small amount of quantum field theory in curved spacetime that we need, the Unruh effect, and then Hawking's calculation. The remaining lectures branch from there, to other derivations of the temperature (Lecture 4), the spectrum and the evaporation (Lecture 5), the trans-Planckian problem (Lecture 6), entropy and information (Lecture 7, the longest), laboratory analogues (Lecture 8), observational searches (Lecture 9) and open problems (Lecture 10). A student interested mainly in the information problem may go from Lecture 3 directly to Lecture 7; the lectures on experiment and observation need only Lectures 3 and 5.

We assume general relativity at the level of a first graduate course, including the Schwarzschild and Kerr solutions, Kruskal coordinates and Penrose diagrams, and the canonical quantisation of a free field. No quantum field theory in curved spacetime is assumed. A student who is at ease with Kruskal coordinates and with the harmonic oscillator in the number basis has all that the first three lectures require. For technical detail we defer to the standard monographs, Birrell and Davies (1982) and Wald (1994).

1.5 Conventions

We use the metric signature $(-, +, +, +)$ and, except in numerical estimates, units in which $\hbar = c = G = k_B = 1$. In these units the Planck mass and the Planck length are both equal to unity, the Schwarzschild radius is $r_s = 2M$, and Eqs. (1.1) and (1.2) become $T_H = \kappa/2\pi = 1/8\pi M$ and $S_{BH} = A/4$.

Problems

1. Restore units in the Schwarzschild surface gravity $\kappa = 1/4M$ to show that $\kappa = c^4/4GM$, and check that $T_H = \hbar\kappa/2\pi ck_B$ reproduces the second and third forms of Eq. (1.1). Evaluate T_H , in kelvin and as $k_B T_H$ in MeV, for a black hole of mass 10^{12} kg.
2. Find the mass of a Schwarzschild black hole whose Hawking temperature equals the present temperature of the cosmic microwave background, 2.725 K, and compare it with the mass of the Moon, 7.3×10^{22} kg. Is a black hole formed by the collapse of a star currently gaining or losing mass through its exchange of radiation with the background?
3. Assuming that a Schwarzschild black hole obeys $dM = T_H dS$ (in units $\hbar = c = G = k_B = 1$), integrate this relation with $T_H = 1/8\pi M$ to show that $S = 4\pi M^2 = A/4$, taking $S = 0$ at $M = 0$. Evaluate S_{BH}/k_B for one solar mass, and show that the heat capacity dM/dT_H is negative.

4. Derive the relation $\lambda = 8\pi^2 r_s / x$ of Section 1.3 between the wavelength of a quantum of energy $x k_B T_H$ and the Schwarzschild radius. The energy flux of a black body per unit frequency peaks at $x \approx 2.82$. Explain in a few sentences why the resulting wavelength is incompatible with the idea that the emitted particles originate in a layer much thinner than r_s outside the horizon.

Notes and further reading

The reader who wants a single companion to the first half of this course should take the lecture notes of Jacobson (2005), which develop quantum field theory in curved spacetime from the beginning and reach the Hawking effect by a route close to ours. Wald (2001) is the standard review of black hole thermodynamics, and the place to see which of the four laws have been proved and under what assumptions; the short review of Carlip (2014) covers similar ground with more attention to microscopic proposals for the entropy. Page (2005) is strongest on the numbers: emission rates, lifetimes, and the entropy of the radiation.

Two longer works repay reading once the basic calculation is familiar. The report of Brout et al. (1995) treats the derivation in great detail, including backreaction and the role of very high frequencies, and is a good source for the physical interpretation of the partner modes. Helfer (2003) asks, critically and with care, which assumptions of the standard derivation are actually justified; we recommend it to anyone inclined to regard the matter as closed.

Lecture 2 Horizons and quantum fields

Lower a small mass on a string towards the horizon of a Schwarzschild black hole and hold it at fixed radius. The proper acceleration of the mass grows without bound as it approaches the horizon. The force per unit mass exerted by the person holding the far end of the string does not: it tends to $c^4/4GM$, which for a solar-mass black hole is $1.5 \times 10^{13} \text{ m s}^{-2}$. The ratio of the second to the first is the redshift factor $\sqrt{f}$, which vanishes at the horizon, and the same redshift, written as a relation between two time coordinates, is the origin of the Hawking effect. Its quantum consequences can already be seen in flat spacetime, by an observer who accelerates uniformly through the vacuum, and that calculation is the heart of this lecture.

2.1 The Schwarzschild geometry and surface gravity

The Schwarzschild metric is

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r}. \quad (2.1)$$

We introduce the tortoise coordinate $r_* = r + 2M \ln|r/2M - 1|$, which ranges over the whole real line as r ranges from $2M$ to infinity, and the retarded and advanced times $u = t - r_*$ and $v = t + r_*$. The radial part of the metric is then conformally flat, $ds^2 = -f du dv + r^2 d\Omega^2$, and outgoing and ingoing radial rays are the lines of constant u and constant v respectively. The coordinate u diverges on the future horizon, which it therefore does not cover. A coordinate that is regular there is the Kruskal coordinate

$$U = -\frac{1}{\kappa} e^{-\kappa u}, \quad V = \frac{1}{\kappa} e^{\kappa v}, \quad \kappa = \frac{1}{4M}, \quad (2.2)$$

in terms of which $ds^2 = -(2M/r) e^{-r/2M} dU dV + r^2 d\Omega^2$, manifestly regular at $r = 2M$. (To check this, note that $dU dV = e^{2\kappa r_*} du dv$ and that $f e^{-2\kappa r_*} = (2M/r) e^{-r/2M}$.) The future horizon is the surface $U = 0$.

The reader should fix in mind the exponential relation between U , the natural coordinate for an observer falling through the horizon, and u , the natural time for an observer at infinity. A wave of fixed frequency at infinity, $e^{-i\omega u} = (-\kappa U)^{i\omega/\kappa}$, performs infinitely many oscillations between any finite value of U and the horizon: traced back towards $U = 0$, its wavelength in U shrinks exponentially.

The constant κ in Eq. (2.2) is the surface gravity. For a stationary black hole with horizon Killing vector ξ^μ (for Schwarzschild, $\xi = \partial_t$), it is defined on the horizon by

$$\xi^\nu \nabla_\nu \xi^\mu = \kappa \xi^\mu, \quad (2.3)$$

that is, it measures the failure of the Killing time to be an affine parameter along the horizon generators.⁴ For a static metric of the form (2.1) with a simple zero of f at r_h one finds $\kappa = f'(r_h)/2$, which gives $\kappa = 1/4M$ for Schwarzschild.

The physical meaning of κ is the one we began with. A static observer at radius r must accelerate to avoid falling in, with proper acceleration $a(r) = M/(r^2\sqrt{f})$, which diverges at the horizon. The product $\sqrt{f}a$, which is the force per unit mass that an observer at infinity would have to exert to hold the static observer on a massless string, remains finite and tends to κ at the horizon. The surface gravity is the redshifted acceleration of a static observer at the horizon. It should not be confused with the local acceleration there, which is infinite.

2.2 The laws of black hole mechanics

The thermodynamic reading of black holes grew out of classical results obtained between 1969 and 1973. Penrose (1969) showed that energy can be extracted from a rotating black hole by particle processes in its ergoregion, and Penrose and Floyd (1971) observed that the extraction processes they analysed were accompanied by an increase in the area of the horizon. Christodoulou (1970) showed that the mass of a Kerr black hole can be decomposed as

$$M^2 = M_{\text{irr}}^2 + \frac{J^2}{4M_{\text{irr}}^2}, \quad A = 16\pi M_{\text{irr}}^2,$$

where the irreducible mass M_{irr} cannot be decreased by any classical process, and that the reversible transformations are precisely those that leave it unchanged. For an extremal Kerr black hole ($J = M^2$) the irreducible mass is $M/\sqrt{2}$, so that at most $1 - 1/\sqrt{2} \approx 29\%$ of the mass can be extracted. Hawking (1971a) then proved that, for matter satisfying the null energy condition, the total area of event horizons cannot decrease.

Bardeen et al. (1973) collected these results into four laws, which are compared with the laws of thermodynamics in Table 1. For a Kerr–Newman black hole the first law reads

$$dM = \frac{\kappa}{8\pi} dA + \Omega_H dJ + \Phi_H dQ, \quad (2.4)$$

where Ω_H and Φ_H are the angular velocity and electrostatic potential of the horizon. The Schwarzschild case is a one-line check: $A = 16\pi M^2$ gives $dA = 32\pi M dM$, and with $\kappa = 1/4M$ the first term of Eq. (2.4) is exactly dM . The correspondence $\kappa \leftrightarrow T$ and $A \leftrightarrow S$ is evident, up to constants that classical physics cannot fix.

Bekenstein (1972, 1973) argued that the analogy reflects a genuine physical identity, and his argument is short enough to give in full. Drop a box containing some entropy into a black hole. For an external observer the entropy has left the accessible universe, and the second law

⁴Since ξ may be rescaled by a constant, so may κ ; the normalisation is fixed by $\xi^\mu \xi_\mu \rightarrow -1$ at infinity. An equivalent definition, often more convenient in calculations, is $\kappa^2 = -\frac{1}{2}(\nabla^\mu \xi^\nu)(\nabla_\mu \xi_\nu)$ evaluated on the horizon.

Table 1: The four laws of black hole mechanics (Bardeen et al., 1973) and the laws of thermodynamics they resemble.

Law	Thermodynamics	Black hole mechanics
Zeroth	The temperature T is uniform in equilibrium	The surface gravity κ is constant on the horizon of a stationary black hole
First	$dE = T dS + \text{work terms}$	$dM = (\kappa/8\pi) dA + \Omega_H dJ + \Phi_H dQ$
Second	$dS \geq 0$ for an isolated system	$dA \geq 0$ in classical processes
Third	$T = 0$ cannot be reached in finitely many steps	$\kappa = 0$ cannot be reached in finitely many steps

of thermodynamics is violated unless the black hole itself carries entropy that increases by at least the amount lost. The area is the only quantity associated with the black hole that never decreases, so Bekenstein proposed that the black hole entropy is proportional to the area, and conjectured the generalised second law: the sum of the black hole entropy and the ordinary entropy outside black holes never decreases (Bekenstein, 1974). Information-theoretic arguments gave him an estimate of the proportionality constant in Planck units, but not a derivation of it.

This is where matters stood in 1973, with the objection of Section 1.1 unanswered: Bardeen et al. (1973) emphasised that a classical black hole, which absorbs but never emits, has zero temperature. To see why the analogy is more than formal, we need quantum field theory.

2.3 Quantum fields in curved spacetime

We work with a free, real, massless scalar field ϕ obeying $\square\phi = 0$ on a globally hyperbolic spacetime. Almost all of the essential physics is already present in this simplest case. The space of classical solutions carries the Klein–Gordon inner product

$$(f, g) = i \int_{\Sigma} d\Sigma n^{\mu} (f^* \partial_{\mu} g - g \partial_{\mu} f^*), \quad (2.5)$$

evaluated on any Cauchy surface Σ with future-directed unit normal n^{μ} , and independent of the choice of Σ .⁵ This inner product is not positive definite: if $(f, f) > 0$ then $(f^*, f^*) < 0$. To quantise the field we choose a complete set of solutions $\{f_i\}$ with positive norm, $(f_i, f_j) = \delta_{ij}$, such that $\{f_i, f_i^*\}$ spans the space of solutions, and we expand

$$\phi = \sum_i (a_i f_i + a_i^{\dagger} f_i^*), \quad [a_i, a_j^{\dagger}] = \delta_{ij}. \quad (2.6)$$

The vacuum associated with this choice is the state $|0_f\rangle$ annihilated by all the a_i , and the $a_i^{\dagger}$ create particles. In Minkowski spacetime there is a preferred choice: the modes that are positive frequency with respect to inertial time, $f \propto e^{-i\omega t}$ with $\omega > 0$, and this choice is the same for all inertial observers. A general curved spacetime has no preferred time coordinate, and different choices of positive-frequency modes lead to inequivalent notions of particle and of vacuum (Fulling, 1973). This ambiguity is not a technical nuisance. It is the central feature of the subject.

The tool for comparing two choices is the Bogoliubov transformation. The reader who has not met it before should work Problem 2 at the end of this lecture before going on; it is the single-mode version of everything that follows. Suppose that a second complete set of

⁵With this sign convention a mode proportional to $e^{-i\omega t}$ with $\omega > 0$ has positive norm in Minkowski spacetime. Some texts use the opposite overall sign.

positive-norm modes $\{p_j\}$ is given. Since both sets span the space of solutions, the new modes can be expanded in terms of the old ones,

$$p_j = \sum_i (\alpha_{ji} f_i + \beta_{ji} f_i^*), \quad (2.7)$$

and the corresponding annihilation operators are related by

$$b_j = (p_j, \phi) = \sum_i (\alpha_{ji}^* a_i - \beta_{ji}^* a_i^\dagger). \quad (2.8)$$

The minus sign comes from the negative norm of f_i^* , and it is easy to lose. The coefficients α_{ji} and β_{ji} are the Bogoliubov coefficients, and orthonormality of the p_j requires $\sum_i (\alpha_{ji} \alpha_{ki}^* - \beta_{ji} \beta_{ki}^*) = \delta_{jk}$. Whenever the β_{ji} are non-zero, the vacuum of one set of modes is not the vacuum of the other. Acting with b_j on $|0_f\rangle$ leaves only the $a_i^\dagger$ terms, so the expected number of p -particles in the f -vacuum is

$$\langle N_j \rangle = \langle 0_f | b_j^\dagger b_j | 0_f \rangle = \sum_i |\beta_{ji}|^2. \quad (2.9)$$

In a spacetime that is asymptotically static in the past and in the future, one can define “in” modes that are positive frequency in the past and “out” modes that are positive frequency in the future. If the field starts in the in-vacuum, Eq. (2.9) gives the number of particles observed at late times. The whole of the Hawking calculation consists in evaluating the β coefficients for a black hole formed by collapse. We first evaluate them for a simpler problem in which the same mechanism operates in flat spacetime.

2.4 A warm-up: the Unruh effect

Take two-dimensional Minkowski spacetime, $ds^2 = -dT^2 + dX^2$, and the right Rindler wedge $X > |T|$. We introduce coordinates (η, ξ) by

$$T = \frac{1}{a} e^{a\xi} \sinh(a\eta), \quad X = \frac{1}{a} e^{a\xi} \cosh(a\eta), \quad (2.10)$$

in terms of which $ds^2 = e^{2a\xi} (-d\eta^2 + d\xi^2)$. The worldline $\xi = 0$ is that of an observer with constant proper acceleration a and proper time η , and the boost Killing vector ∂_η generates the observer’s time translations. The lines $T = \pm X$ are horizons for this observer: signals from beyond $T = X$ can never reach it. In terms of the null coordinates $U = T - X$ and $u = \eta - \xi$, and $V = T + X$ and $v = \eta + \xi$, the transformation (2.10) becomes

$$U = -\frac{1}{a} e^{-au}, \quad V = \frac{1}{a} e^{av}. \quad (2.11)$$

This is Eq. (2.2) with a in place of κ . The accelerated observer uses u as a time coordinate; the inertial observer uses U .

A massless field in two dimensions splits into right-moving and left-moving parts, and it suffices to consider the right-moving part, which depends only on U . A mode that is positive frequency for the accelerated observer is $e^{-i\omega u}$ with $\omega > 0$, which by Eq. (2.11) can be written as

$$e^{-i\omega u} = (-aU)^{i\omega/a}, \quad U < 0. \quad (2.12)$$

We want to know how much of this mode is positive frequency for inertial observers. The direct route is a Fourier transform in U , which leads to Gamma functions; a shorter one, due to Unruh (1976), uses analyticity. A function of U is a superposition of inertial positive-frequency modes $e^{-i\Omega U}$, $\Omega > 0$, if and only if it is analytic and bounded in the lower half of the complex U -plane,

since each $e^{-i\Omega U}$ decays there. The function (2.12) is defined only for $U < 0$, and we continue it to $U > 0$ through the lower half-plane. Writing $U = R e^{i\varphi}$ with φ running from $-\pi$ to 0, we see that $-U = R e^{i(\varphi+\pi)}$ acquires the phase $e^{i\pi}$ when U reaches the positive real axis, so that

$$(-aU)^{i\omega/a} \longrightarrow e^{i\pi \cdot i\omega/a} (aU)^{i\omega/a} = e^{-\pi\omega/a} (aU)^{i\omega/a}, \quad U > 0. \quad (2.13)$$

The function equal to $(-aU)^{i\omega/a}$ for $U < 0$ and to $e^{-\pi\omega/a} (aU)^{i\omega/a}$ for $U > 0$ is therefore purely positive frequency with respect to inertial time. This is the cleanest derivation of thermality that we know, and the one we recommend remembering: Lecture 3 repeats it with κ in place of a .

Remark 2.1. A common error is to continue through the upper half-plane. That gives the factor $e^{+\pi\omega/a}$, a purely negative-frequency function, and in the end a negative “occupation number”. The choice of half-plane is fixed by the convention $e^{-i\Omega U}$ for positive frequency.

For $U > 0$, that is in the left Rindler wedge, the function $(aU)^{i\omega/a}$ is the complex conjugate of a mode that is positive frequency with respect to the future-directed boost time of that wedge. Translated into operators, this means that the Minkowski vacuum $|0_M\rangle$ is annihilated by the combinations

$$\left(b_\omega^R - e^{-\pi\omega/a} b_\omega^{L\dagger}\right)|0_M\rangle = 0, \quad \left(b_\omega^L - e^{-\pi\omega/a} b_\omega^{R\dagger}\right)|0_M\rangle = 0, \quad (2.14)$$

where b_ω^R and b_ω^L annihilate Rindler quanta in the right and left wedges. The number of Rindler quanta seen by the accelerated observer now takes two lines. Write $x = e^{-\pi\omega/a}$ and $N = \langle 0_M | b_\omega^{R\dagger} b_\omega^R | 0_M \rangle$. By Eq. (2.14), $N = x^2 \langle 0_M | b_\omega^L b_\omega^{L\dagger} | 0_M \rangle = x^2(1 + N)$, where we have used the symmetry between the two wedges. Hence

$$N = \frac{x^2}{1 - x^2} = \frac{1}{e^{2\pi\omega/a} - 1}, \quad (2.15)$$

which is a Planck distribution at the Unruh temperature

$$T_U = \frac{a}{2\pi} = \frac{\hbar a}{2\pi c k_B}. \quad (2.16)$$

An accelerated observer in the Minkowski vacuum perceives a thermal bath. For accelerations of everyday magnitude the effect is absurdly small: $a = 9.8 \text{ m s}^{-2}$ gives $T_U \approx 4 \times 10^{-20} \text{ K}$, and a temperature of 1 K requires $a \approx 2.5 \times 10^{20} \text{ m s}^{-2}$.

The Minkowski vacuum can be written explicitly in terms of Rindler states. One verifies directly (Problem 4) that the state

$$|0_M\rangle = \prod_\omega \sqrt{1 - e^{-2\pi\omega/a}} \sum_{n=0}^{\infty} e^{-\pi n\omega/a} |n_\omega\rangle_L \otimes |n_\omega\rangle_R \quad (2.17)$$

satisfies Eq. (2.14). The Minkowski vacuum is therefore a pure, entangled state of the two wedges. An observer confined to the right wedge has no access to the left one, and describes the field by the reduced density matrix obtained by tracing over the left wedge, $\rho_R = \prod_\omega (1 - e^{-\omega/T_U}) \sum_n e^{-n\omega/T_U} |n_\omega\rangle \langle n_\omega|$, which is exactly thermal. States of the form (2.17) are known as thermofield-double states. The thermality is a consequence of entanglement across the horizon, and this point will be central to the information problem in Lecture 7.

The thermal response of a uniformly accelerated detector in the Minkowski vacuum was found by Unruh (1976), following the work of Fulling (1973) on the non-uniqueness of the vacuum and of Davies (1975) on the analogy between Rindler and Schwarzschild particle production.⁶ The same paper introduced the model particle detector that now bears Unruh’s name. The large literature on the Unruh effect has been reviewed by Crispino et al. (2008).

⁶For this reason the effect is sometimes called the Fulling–Davies–Unruh effect.

2.5 Superradiance

A second precursor of Hawking's result came from the scattering of waves by rotating bodies. Zel'dovich (1971) showed that a rotating absorbing cylinder amplifies incident waves of frequency ω and azimuthal number m whenever

$$0 < \omega < m \Omega, \quad (2.18)$$

where Ω is the angular velocity of the body. The origin of the effect is simple. In the frame co-rotating with the body the wave has frequency $\omega - m\Omega$, which is negative when Eq. (2.18) holds, so absorption in the co-rotating frame is emission in the frame of the observer. Zel'dovich argued that in quantum theory the same body should also emit spontaneously in these "superradiant" modes. Starobinsky (1973) computed the corresponding amplification for a Kerr black hole, for which Ω is replaced by the angular velocity of the horizon Ω_H ; Teukolsky and Press (1974) extended the analysis to electromagnetic and gravitational waves, and Unruh (1974) confirmed, by quantising the field on the Kerr background, that a rotating black hole emits spontaneously in the superradiant modes. For a black hole the condition (2.18) also follows from the first and second laws alone (Problem 5).

The reasoning of Zel'dovich and Starobinsky was an important motivation for Hawking's investigation. What it did not prepare anyone for was the answer: emission in all modes, persisting when the black hole does not rotate at all. Its later applications, from black hole instabilities to searches for ultralight bosons, are reviewed by Brito et al. (2020).

Problems

1. The Reissner–Nordström metric has the form (2.1) with $f(r) = 1 - 2M/r + Q^2/r^2$, $|Q| < M$. Show that $f(r) = (r - r_+)(r - r_-)/r^2$ with $r_{\pm} = M \pm \sqrt{M^2 - Q^2}$, and that the surface gravity of the outer horizon is $\kappa = (r_+ - r_-)/2r_+^2$. Show that $\kappa \rightarrow 0$ in the extremal limit $|Q| \rightarrow M$, and that $\kappa = 1/4M$ when $Q = 0$.
2. Let a be the annihilation operator of a single oscillator with vacuum $|0_a\rangle$, and define $b = a \cosh r + a^\dagger \sinh r$ with r real. Show that $[b, b^\dagger] = 1$, that $\langle 0_a | b^\dagger b | 0_a \rangle = \sinh^2 r$, and that the state annihilated by b is proportional to $\exp(-\frac{1}{2} \tanh r (a^\dagger)^2) |0_a\rangle$. Identify α and β by comparison with Eq. (2.8), and check the normalisation condition $|\alpha|^2 - |\beta|^2 = 1$. (Hint: a acts on a function of $a^\dagger$ as $\partial/\partial a^\dagger$.)
3. Using Eq. (2.16), evaluate T_U for $a = 9.8 \text{ m s}^{-2}$ and find the acceleration for which $T_U = 1 \text{ K}$. Then evaluate the surface gravity $c^4/4GM$ of a solar-mass black hole in m s^{-2} , and show that inserting it in place of a in Eq. (2.16) reproduces the temperature $6.2 \times 10^{-8} \text{ K}$ of Eq. (1.1).
4. Show that the state (2.17) is normalised and satisfies both conditions (2.14). For a single mode of frequency ω , show that the entanglement entropy of the right wedge, $-\text{tr } \rho_R \ln \rho_R$, equals $-\ln(1 - e^{-\omega/T_U}) + (\omega/T_U)/(e^{\omega/T_U} - 1)$, the thermal entropy of an oscillator. Show that it tends to zero for $\omega \gg T_U$ and diverges logarithmically as $\omega/T_U \rightarrow 0$.
5. A wave of frequency $\omega > 0$ and azimuthal number m carries angular momentum and energy in the ratio $dJ/dM = m/\omega$. Apply the first law (2.4) with $dQ = 0$ to the absorption of such a wave by a Kerr black hole, and use the area theorem $dA \geq 0$ to show that the black hole can lose mass to the wave, that is, the wave can be amplified, only if $\omega < m \Omega_H$.

Notes and further reading

For the field theory of this lecture, Birrell and Davies (1982) remains the standard reference on mode sums, Bogoliubov transformations and particle detectors; it is the book to consult when a factor of 2π goes missing. Wald (1994) is more careful about what a quantum state on a curved spacetime is, and its treatment of the Unruh effect does not rely on the formal infinite product of Eq. (2.17), which does not define a vector in the Fock space of Rindler quanta. Students meeting the subject for the first time will find the gentlest route in the lecture notes of Jacobson (2005), which are close in spirit to the treatment given here. The laws of black hole mechanics, with an account of which have been proved and in what generality, are reviewed by Wald (2001).

Two reviews cover the precursors in depth. Crispino et al. (2008) discuss the Unruh effect from the points of view of detectors and of field theory, together with its applications and the proposals for observing it. Brito et al. (2020) is the standard modern account of superradiance, from Zel'dovich's cylinder to the bounds on ultralight bosons from the spins of astrophysical black holes.

Lecture 3 Hawking's calculation

In 1973 Hawking set out to put the argument of Zel'dovich and Starobinsky (Section 2.5) on a firm footing. He expected to confirm that a rotating black hole emits in its superradiant modes, and that the emission stops when the rotation does. It does not stop. A Schwarzschild black hole formed by collapse and then left alone emits a steady flux of particles with a thermal spectrum at the temperature $\kappa/2\pi$ (Hawking, 1974, 1975).⁷ We follow his argument in a stripped-down form that keeps only the steps that matter.

3.1 Set-up

We take a spherical star that collapses to form a Schwarzschild black hole, and on the resulting spacetime (Fig. 1) we place a free massless scalar field. We assume that before the collapse the field is in the vacuum defined by the natural time coordinate at past null infinity $\mathcal{I}^-$: nothing comes in from infinity. We assume that the collapsing matter is transparent to the field, so that an ingoing wave passes through the centre and comes out again; this is a convenience, and nothing essential depends on it. And we neglect the backreaction of the field on the geometry. The last assumption is the one that eventually fails, and we return to it in Section 5.6.

The question is what an observer at future null infinity $\mathcal{I}^+$ detects at late times. In the language of Section 2.3, the in-modes are positive frequency with respect to the advanced time v on $\mathcal{I}^-$, the out-modes with respect to the retarded time u on $\mathcal{I}^+$, and the number of out-quanta in the in-vacuum is $\sum |\beta|^2$ by Eq. (2.9). A single property of the geometry turns out to fix the β coefficients.

3.2 The exponential redshift

Consider an ingoing radial light ray that leaves $\mathcal{I}^-$ at advanced time v . It falls inwards, passes through the centre of the star and emerges as an outgoing ray. There is a last ray that escapes to infinity; we call its advanced time v_0 . This ray generates the event horizon. A ray that leaves slightly before v_0 lingers just outside the horizon and reaches $\mathcal{I}^+$ at a late retarded time u (the thick ray in Fig. 1). We want $u(v)$ as $v \rightarrow v_0$.

⁷The announcement was a two-page letter to *Nature* in March 1974. The detailed paper, submitted the same year, appeared in 1975.

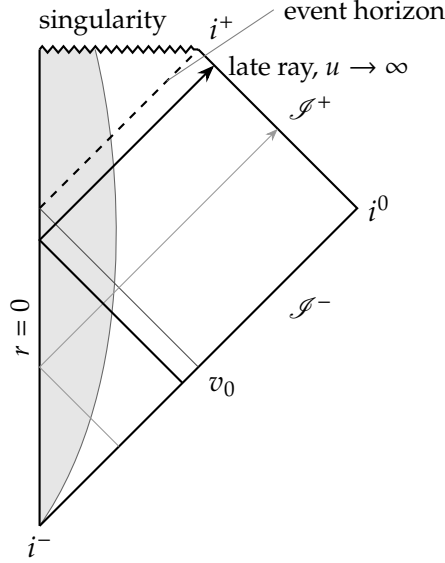


Figure 1: Penrose diagram of a black hole formed by the collapse of a star (shaded). Ingoing rays from $\mathcal{J}^-$ pass through the centre $r = 0$ and emerge as outgoing rays. The ray that leaves $\mathcal{J}^-$ at the advanced time v_0 generates the event horizon (dashed). A ray that leaves $\mathcal{J}^-$ slightly before v_0 (thick) stays close to the horizon for a long time and reaches $\mathcal{J}^+$ at a late retarded time u ; an earlier ray (grey) escapes promptly.

The trick is to label the outgoing rays near the horizon not by u , which is infinite on the horizon, but by the Kruskal coordinate U of Eq. (2.2), which is regular there and vanishes on the horizon itself. The map $v \mapsto U(v)$ is fixed by the passage of the ray through the star. Whatever the star is made of, this passage is smooth, so $U(v)$ is smooth and monotonic with $U(v_0) = 0$, and near v_0 the first term of its Taylor series is enough:

$$U \simeq -c(v_0 - v), \quad c > 0. \quad (3.1)$$

Outside the star, the relation between U and u is pure Schwarzschild geometry, $U = -\kappa^{-1}e^{-\kappa u}$. Eliminating U gives

$$v_0 - v \simeq C e^{-\kappa u}, \quad u \rightarrow \infty, \quad (3.2)$$

with $C = (c\kappa)^{-1}$. This is the exponential relation promised in Section 2.1: a wave that reaches $\mathcal{J}^+$ with frequency ω at retarded time u left $\mathcal{J}^-$ with a frequency of order $\omega e^{\kappa u}$, a redshift that grows exponentially with time.

A common error at this point is to look for the exponential inside the star. It is not there. The interior supplies the linear map (3.1) and fixes only the constant c ; the exponential comes from the exterior relation between U and u , at a rate set by the surface gravity alone.

The numbers are worth seeing once. For a solar-mass black hole, $\kappa^{-1} = 4GM/c^3 \approx 20 \mu\text{s}$. A millisecond of retarded time therefore multiplies the redshift by about $e^{50} \approx 5 \times 10^{21}$. A Hawking quantum detected a couple of milliseconds after the horizon forms must be traced back to a wave on $\mathcal{J}^-$ with a frequency above the Planck scale (Problem 3). This is the trans-Planckian problem of Lecture 6.

Remark 3.1. The details of the collapse enter Eq. (3.2) only through the constant C , while the exponent is fixed by the surface gravity of the final black hole. We should therefore expect the late-time radiation not to depend on how the black hole was formed, and the argument to apply to any horizon, gravitational or not, across which the natural coordinates of the two sides are related exponentially. Both expectations are borne out (Section 4.5 and Lecture 8).

3.3 The Bogoliubov coefficients

Take an out-mode of frequency ω and angular quantum numbers (ℓ, m) , which near $\mathcal{S}^+$ has the form $p_\omega \propto r^{-1} e^{-i\omega u} Y_{\ell m}$, and run it backwards in time. Part of it scatters off the curvature potential outside the star and reaches $\mathcal{S}^-$ at late advanced times as $e^{-i\omega v}$, which is positive frequency and contributes nothing to β . The rest, a fraction $\Gamma_{\omega\ell}$ in probability, enters the star, passes through the centre and reaches $\mathcal{S}^-$ just before v_0 . Substituting $u(v)$ from Eq. (3.2) into $e^{-i\omega u}$, we find

$$p_\omega|_{\mathcal{S}^-} \propto \begin{cases} \exp\left[\frac{i\omega}{\kappa} \ln \frac{v_0 - v}{C}\right], & v < v_0, \\ 0, & v > v_0. \end{cases} \quad (3.3)$$

This function oscillates infinitely often as $v \rightarrow v_0$ and vanishes beyond it. It is not positive frequency with respect to v , and we need to know by how much it fails to be.

Rather than compute Fourier integrals, we repeat the analyticity argument of Section 2.4 with $v_0 - v$ in place of $-U$ and κ in place of a . A function of v is a superposition of in-modes $e^{-i\omega'v}$, $\omega' > 0$, if and only if it is analytic and bounded in the lower half v -plane. Continuing $(v_0 - v)^{i\omega/\kappa}$ to $v > v_0$ through that half-plane rotates $v_0 - v$ by $e^{i\pi}$ and produces the factor $e^{-\pi\omega/\kappa}$, as in Eq. (2.13). Hence (3.3) plus $e^{-\pi\omega/\kappa}$ times its mirror image, the function $[(v - v_0)/C]^{i\omega/\kappa}$ supported on $v > v_0$, is purely positive frequency. Its negative-frequency content vanishes, so the negative-frequency content of the traced mode equals $-e^{-\pi\omega/\kappa}$ times that of the mirror image; and the reflection $v - v_0 \rightarrow v_0 - v$ shows that the negative-frequency content of the mirror image has the modulus of the positive-frequency content of the traced mode. Together these give

$$|\beta_{\omega\omega'}|^2 = e^{-2\pi\omega/\kappa} |\alpha_{\omega\omega'}|^2 \quad (3.4)$$

for the transmitted part of the mode.⁸ Hawking (1975) obtained the same relation by evaluating the integrals explicitly. Both are proportional to the same Gamma function, and their ratio comes entirely from the phase of $\pm i\omega'$ raised to a complex power (Problem 2).

The rest is algebra. The normalisation condition of Section 2.3, applied to the transmitted part of the mode, reads $\sum_{\omega'} (|\alpha_{\omega\omega'}|^2 - |\beta_{\omega\omega'}|^2) = \Gamma_{\omega\ell}$. With Eq. (3.4) it becomes $(e^{2\pi\omega/\kappa} - 1) \sum_{\omega'} |\beta_{\omega\omega'}|^2 = \Gamma_{\omega\ell}$, so that

$$\langle N_{\omega\ell m} \rangle = \sum_{\omega'} |\beta_{\omega\omega'}|^2 = \frac{\Gamma_{\omega\ell}}{e^{2\pi\omega/\kappa} - 1}. \quad (3.5)$$

For modes of sharp frequency this number carries a divergent factor, because such a mode lasts forever and so does the emission. With normalised wave packets the divergence becomes the duration of the emission, and one obtains a steady rate; the factor $1/2\pi$ below counts wave packets per unit time and per unit frequency. For a Kerr–Newman black hole the general result is

$$\frac{dN}{dt d\omega} = \frac{1}{2\pi} \sum_{\ell, m} \frac{\Gamma_{\omega\ell m}}{\exp[(\omega - m\Omega_H - q\Phi_H)/T_H] \mp 1}, \quad T_H = \frac{\kappa}{2\pi}, \quad (3.6)$$

where q is the charge of the emitted particle, the upper sign applies to bosons and the lower sign to fermions. With outer and inner horizons at $r_\pm$ and $a = J/M$, the surface gravity and the angular velocity of the horizon are $\kappa = (r_+ - r_-)/[2(r_+^2 + a^2)]$ and $\Omega_H = a/(r_+^2 + a^2)$; for $r_- = a = 0$ the first reduces to $1/4M$, as it should. Superradiance (2.18) is built in: for bosons with $\omega < m\Omega_H + q\Phi_H$ the Planck factor is negative, and so is the greybody factor, since such waves are amplified rather than absorbed. The positive product is the spontaneous emission that Zel'dovich and Starobinsky anticipated.

⁸References differ in whether α and β carry complex conjugates and in the order of their indices. Only moduli enter here, so these conventions do not affect any result of this lecture.

Equation (3.6) is the central result of the course. A black hole emits particles of every species as a black body at temperature T_H , except that the Planck spectrum is multiplied by the greybody factor $\Gamma_{\omega\ell m}$. This is the probability that a wave leaving the vicinity of the horizon crosses the curvature potential and reaches infinity. By time-reversal symmetry it equals the probability that a wave sent in from infinity is absorbed, and it is computed in Section 5.1.

3.4 A heuristic derivation from the Unruh effect

The similarity with the Unruh effect of Section 2.4 is no accident, and it yields the temperature in a few lines. Close to the horizon, over distances small compared with M , the Schwarzschild geometry is indistinguishable from flat spacetime in Rindler coordinates. A static observer at radius r is then simply an accelerated observer, with proper acceleration $a(r) = M/(r^2\sqrt{f})$. Suppose that the field near the horizon looks like the vacuum to freely falling observers, as it must if nothing singular happens there. The static observer then sees a thermal bath at the local Unruh temperature $a(r)/2\pi$. Thermal radiation that climbs out of a gravitational potential is redshifted, and the temperature measured at infinity is lower by the factor $\sqrt{f}$ (the Tolman relation; see Wald, 1994).⁹ Hence

$$T_\infty = \sqrt{f(r)} \frac{a(r)}{2\pi} = \frac{M}{2\pi r^2} \xrightarrow{r \rightarrow 2M} \frac{1}{8\pi M} = \frac{\kappa}{2\pi}, \quad (3.7)$$

in agreement with Eq. (3.6). The limit $r \rightarrow 2M$ is essential: only there is the local state close to the Rindler vacuum, so only there is the local temperature the Unruh temperature.

For quick reasoning about horizons this is the most useful form of the result. It shows that the Hawking temperature is the Unruh temperature of the surface gravity, and it isolates the assumption on which everything rests: the state must look locally like the vacuum to an infalling observer at the horizon. Its limitations are as instructive. It gives no greybody factors, and it cannot tell an outgoing flux from equilibrium with an incoming bath, since both are regular on the future horizon (Section 4.1). Hawking's calculation derives the regularity from the collapse instead of assuming it, and selects the outgoing flux.

3.5 The state of the radiation

We now ask for the late-time state itself, not only for the number of quanta. The analyticity argument of Section 3.3 combined each out-mode with its mirror image on $v > v_0$. Followed forwards in time, the mirror image is a mode that travels into the black hole; we call it the partner mode. Exactly as in Eq. (2.17), the in-vacuum can be written, for each late-time mode, as

$$|0_{\text{in}}\rangle \propto \sum_{n=0}^{\infty} e^{-\pi n \omega / \kappa} |n_\omega\rangle_{\text{out}} \otimes |n_\omega\rangle_{\text{partner}}. \quad (3.8)$$

The state is pure. Each outgoing quantum, however, is entangled with a partner quantum behind the horizon. An observer outside has no access to the partners and must describe the radiation by the reduced density matrix obtained by tracing them out, $\rho_\omega = (1 - e^{-\omega/T_H}) \sum_n e^{-n\omega/T_H} |n_\omega\rangle\langle n_\omega|$, which is exactly thermal. Wald (1975) and, independently, Parker (1975) proved that the radiation is described by a thermal density matrix with no correlations between different out-modes.

Equation (3.8) is the precise version of the pair picture of Section 1.3. It also contains the seed of the information problem. If the black hole eventually disappears, the partners disappear with it, and the radiation left behind is in a mixed state (Lecture 7).

⁹Tolman's result is that $T\sqrt{-g_{tt}}$ is constant in a static spacetime in thermal equilibrium.

Remark 3.2. The pair picture invites the misreading that each quantum is created at the horizon, close to its partner. The modes in Eq. (3.8) are not localised in this way: a typical Hawking quantum, with a frequency of a few T_H , has a wavelength of some tens of Schwarzschild radii (Remark 5.1).

3.6 Energy, area, and entropy

Positive energy carried to infinity must be paid for by a loss of mass. The outgoing flux at infinity is accompanied by a flux of negative energy across the horizon (Section 4.3), and the area decreases. This does not contradict the area theorem of Hawking (1971a), because the renormalised stress-energy tensor violates the null energy condition near the horizon. It does mean that the area theorem is not fundamental. Its place is taken by the generalised second law, which survives because the entropy of the radiation more than compensates the loss of black hole entropy (Bekenstein, 1974; Wald, 2001).

Once the temperature is known, the first law (2.4) fixes the entropy. For a Schwarzschild black hole $dS = dM/T_H = 8\pi M dM$, which integrates to $S = 4\pi M^2 = A/4$, the Bekenstein–Hawking entropy (1.2). The constant of integration is set to zero by requiring $S \rightarrow 0$ as $M \rightarrow 0$, an assumption that the semiclassical calculation cannot check. The resulting thermodynamics is peculiar. Since $T_H \propto 1/M$, the heat capacity

$$C = \frac{dM}{dT_H} = -8\pi M^2 \quad (3.9)$$

is negative. A black hole that radiates becomes hotter and radiates faster, and one in contact with an infinite heat bath cannot be in stable equilibrium: slightly hotter than the bath, it evaporates; slightly colder, it grows. The same instability reappears in the Euclidean approach (Section 4.2).

3.7 Orders of magnitude

Restoring units, the Hawking temperature reads $T_H \simeq 6.17 \times 10^{-8} \text{ K } (M_\odot/M)$, or, in energy units, $k_B T_H \simeq 1.06 \text{ GeV } (10^{13} \text{ g}/M)$. Table 2 gives representative values; the lifetimes are derived in Lecture 5.

Two conclusions follow at once. Astrophysical black holes are far colder than the cosmic microwave background, whose temperature is 2.725 K: setting $T_H = 2.725 \text{ K}$ in the first formula gives $M \simeq 2.3 \times 10^{-8} M_\odot \simeq 4.5 \times 10^{22} \text{ kg}$, somewhat less than the mass of the Moon. Every black hole formed by stellar collapse therefore absorbs more radiation today than it emits. At the other end, black holes lighter than about 10^{16} g radiate at nuclear and particle-physics energies, and those near $5 \times 10^{14} \text{ g}$ would be completing their evaporation today. Only the early universe could have made black holes this light, which is why the search for Hawking radiation is a search for primordial black holes (Lecture 9).

Problems

1. *Surface gravity of a static horizon.* For the metric $ds^2 = -f(r) dt^2 + dr^2/f(r) + r^2 d\Omega^2$ with a simple zero of f at r_h , show that near the horizon the tortoise coordinate behaves as $r_* \simeq (2\kappa)^{-1} \ln|r - r_h|$ with $\kappa = f'(r_h)/2$, and that the radial metric $-f du dv$ is regular at $r = r_h$ when written in terms of $U = -\kappa^{-1} e^{-\kappa u}$ and $V = \kappa^{-1} e^{\kappa v}$, so that Eq. (3.2) holds with this κ . Apply the result to the Reissner–Nordström metric, $f = (1 - r_+/r)(1 - r_-/r)$, and show that $T_H = (r_+ - r_-)/(4\pi r_+^2)$, in agreement with the Kerr–Newman formula at $a = 0$.
2. *The ratio from the Fourier integrals.* Up to a common factor, $\alpha_{\omega\omega'}$ and $\beta_{\omega\omega'}$ for the transmitted mode (3.3) are given by $I_\pm = \int_0^\infty dx x^{i\omega/\kappa} e^{\mp i\omega'x - \epsilon x}$ with $x = v_0 - v$ and $\epsilon \rightarrow 0^+$ (the upper

Table 2: Characteristic scales of Schwarzschild black holes. The lifetimes are order-of-magnitude estimates that depend on the particle species emitted (Section 5.2); they neglect accretion, including the absorption of the cosmic microwave background.

Mass	Example	$r_s = 2GM/c^2$	T_H	Lifetime
$10 M_\odot$	Stellar black hole	30 km	6×10^{-9} K	$\sim 10^{70}$ yr
$M_\odot$		3 km	6×10^{-8} K	$\sim 10^{67}$ yr
4.5×10^{25} g	$T_H = T_{\text{CMB}}$	0.07 mm	2.7 K	$\sim 10^{44}$ yr
10^{20} g	Asteroid mass	1.5×10^{-10} m	≈ 100 eV	$\sim 10^{26}$ yr
5×10^{14} g	Evaporating today	7×10^{-16} m	≈ 20 MeV	$\approx 1.4 \times 10^{10}$ yr
10^9 g		1.5×10^{-21} m	≈ 10 TeV	~ 0.4 s

sign for α). Using $\int_0^\infty dx x^{s-1} e^{-px} = \Gamma(s) p^{-s}$ for $\text{Re } p > 0$, with the principal branch of p^{-s} , show that $|I_+/I_-| = e^{\pi\omega/\kappa}$, and hence recover Eq. (3.4).

3. *How soon does the trans-Planckian problem appear?* Working to logarithmic accuracy, show that a quantum of frequency $\omega \sim T_H$ that reaches $\mathcal{I}^+$ at retarded time u (measured from the formation of the horizon) left $\mathcal{I}^-$ with a frequency above the Planck mass once $u \gtrsim 4M \ln(8\pi M/m_P)$. Evaluate this time in seconds for $M = M_\odot$ and for $M = 5 \times 10^{14}$ g. (Answers: about 2 ms and about 2×10^{-22} s.)
4. *The local temperature near the horizon.* Write $r = 2M + \rho^2/8M$, so that ρ is the proper distance from the horizon for $\rho \ll M$. Show that the local temperature $T_H/\sqrt{f}$ seen by a static observer is $1/(2\pi\rho)$ to leading order, independent of M , and that this is the Unruh temperature for the proper acceleration of that observer. Explain why this local thermal bath is consistent with an infalling observer seeing no particles.
5. *Negative heat capacity.* Show that for a Schwarzschild black hole $C = -2S_{\text{BH}}$, and evaluate C in J K^{-1} for $M = M_\odot$ (answer: about $-3 \times 10^{54} \text{ J K}^{-1}$). A black hole whose mass is slightly larger than the value in Table 2 for which $T_H = T_{\text{CMB}}$ sits in the cosmic microwave background. Ignoring the expansion of the universe and all accretion other than of the background radiation, state whether it grows or shrinks, and whether it moves towards or away from the equilibrium mass.

Notes and further reading

The original paper, Hawking (1975), is still worth reading in full once the argument of this lecture is familiar; its treatment of wave packets and of the scattering outside the star is more careful than ours, and the short announcement Hawking (1974) shows how quickly the main points were in place. For a first reading we recommend Jacobson (2005), which follows the same route as this lecture with more attention to the near-horizon geometry and to the trans-Planckian question. Birrell and Davies (1982) give the standard textbook account of particle creation by a collapsing body, and Wald (1994) gives the mathematically careful version, including the construction of the state that underlies Eq. (3.8).

On the thermal character of the state, the papers of Wald (1975) and Parker (1975) remain the references; both are short. Brout et al. (1995) is a long but readable primer that is especially good on the partner modes and on the physical interpretation of the pair picture.

Table 3: The three standard states on the Schwarzschild spacetime.

State	Future horizon	Past horizon	Physical situation
Boulware	Singular	Singular	Exterior of a static star; no flux at infinity
Unruh	Regular	Singular	Black hole formed by collapse; outgoing thermal flux at T_H
Hartle–Hawking	Regular	Regular	Black hole in equilibrium with a thermal bath at T_H

Lecture 4 Other routes to the same temperature

Hawking’s calculation uses a collapsing star, geometric optics through its interior, and a vacuum imposed on $\mathscr{I}^-$ long before the black hole exists. None of these appears in the answer, $T_H = \kappa/2\pi$, which depends on the final black hole alone. Over the following decades the effect was rederived in many ways that never mention the star. All give $\kappa/2\pi$. They differ in what they assume and in what they establish, and as a rule the more economical a derivation is, the more it assumes.

4.1 The choice of state

On the maximally extended Schwarzschild spacetime, an eternal black hole, nothing fixes the state: initial data must be given on the past horizon as well as on $\mathscr{I}^-$. Comparing the three standard choices (Table 3) is the best way to see what the collapse selects.

The Boulware state (Boulware, 1975) is the vacuum defined with respect to the Killing time t . It has no radiation at infinity, but its stress-energy tensor diverges on both horizons, as that of the Rindler vacuum of Section 2.4 does on the Rindler horizons. It is the right state outside a static star larger than $2M$, and unphysical for a black hole. The Hartle–Hawking state (Hartle and Hawking, 1976; Israel, 1976) is regular on both horizons and describes a black hole in thermal equilibrium with a bath of radiation at T_H . It is the gravitational analogue of the Minkowski vacuum restricted to a Rindler wedge. The Unruh state (Unruh, 1976) is regular on the future horizon, has nothing coming in from $\mathscr{I}^-$, and carries an outgoing thermal flux at infinity. It reproduces the late-time state produced by collapse, and so describes an evaporating black hole. Candelas (1980) computed the vacuum polarisation in all three.

The table makes the lesson of Section 3.4 precise. Regularity on the future horizon goes with outgoing thermal radiation at T_H ; regularity on the past horizon adds an incoming bath that balances it. A collapsing star has no past horizon, so the singularity of the Unruh state there costs nothing.

Kay and Wald (1991) turned this into a theorem. For spacetimes with a bifurcate Killing horizon they proved that there is at most one stationary state that is invariant under the isometries and satisfies the Hadamard condition (its two-point function has the short-distance singularity of the Minkowski vacuum). When such a state exists, it is a Kubo–Martin–Schwinger (KMS) state, that is, a state of thermal equilibrium, at temperature $\kappa/2\pi$ with respect to the Killing time.¹⁰ Thermality is therefore not an extra assumption: it follows from requiring that the state look like the vacuum at short distances across the horizon. This is, in our view, the cleanest statement of why the answer is thermal. The theorem concerns a stationary state on the eternal black hole; the link with collapse is supplied by Fredenhagen and Haag (1990) (Section 4.5). Kay

¹⁰The condition, due to Kubo and to Martin and Schwinger in the late 1950s, characterises equilibrium by an analyticity property of correlation functions in complex time. It makes sense even when no Gibbs density matrix exists.

and Wald (1991) also showed that no such state exists on the Kerr spacetime, a result related to the presence of superradiant modes.

4.2 Euclidean methods

The quickest route to the temperature is to continue to imaginary time. Set $t = -i\tau$ in the Schwarzschild metric and look at the resulting Riemannian metric near $r = 2M$. With the proper distance from the horizon introduced through $r = 2M + \rho^2/8M$, one finds $f \simeq \rho^2/16M^2$ and $dr^2/f \simeq d\rho^2$, so that

$$ds_E^2 \simeq \rho^2 d(\kappa\tau)^2 + d\rho^2 + 4M^2 d\Omega^2. \quad (4.1)$$

The first two terms are the flat metric in polar coordinates, with $\kappa\tau$ as the polar angle. The geometry is smooth at $\rho = 0$ only if this angle has period 2π ; otherwise there is a conical singularity. The imaginary time must therefore be periodic, $\tau \sim \tau + 2\pi/\kappa$. Since a thermal state at temperature T corresponds to imaginary time of period $1/T$, the black hole comes with the temperature $T = \kappa/2\pi$. Note that τ is normalised at infinity, so the period gives the temperature measured there.

The periodicity is not a new piece of physics. The factor $e^{-\pi\omega/\kappa}$ of Eq. (2.13), with a replaced by κ , is exactly what a positive-frequency mode $e^{-i\omega t}$ acquires when t is shifted by $-i\pi/\kappa$, half the Euclidean period. The analyticity argument of Lectures 2 and 3 and the absence of a conical singularity are the same statement.

Gibbons and Hawking (1977a) went further and read the Euclidean gravitational path integral, with period $\beta = 8\pi M$, as a thermal partition function, $Z = \int \mathcal{D}g e^{-I_E[g]}$. In the saddle-point approximation $Z \simeq e^{-I_E}$, where $I_E = \beta^2/16\pi$ is the on-shell action of the Euclidean Schwarzschild solution.¹¹ The standard thermodynamic relations give

$$E = -\frac{\partial \ln Z}{\partial \beta} = \frac{\beta}{8\pi} = M, \quad S = \beta E + \ln Z = \frac{\beta^2}{16\pi} = 4\pi M^2 = \frac{A}{4}, \quad (4.2)$$

so a purely geometrical calculation returns both the mass and the Bekenstein–Hawking entropy. The same method gives the cosmological horizon of de Sitter space the temperature $H/2\pi$, where H is the Hubble rate (Gibbons and Hawking, 1977b). Earlier, Hartle and Hawking (1976) had obtained the flux by continuing the Feynman propagator in the complexified Schwarzschild geometry, finding that the probabilities of emission and absorption of a particle differ by the factor $e^{-\omega/T_H}$. Israel (1976) observed that the Hartle–Hawking state is a thermofield double of the two exterior regions of the extended geometry, in exact analogy with Eq. (2.17).¹²

What does the Euclidean argument establish? It gives the temperature in three lines and the entropy in five, and no other derivation is as economical. It describes equilibrium, however, not a process, and says nothing about collapse, the direction of the flux or greybody factors. The entropy calculation is the more striking and the less secure of its two results, because the saddle is not a minimum of the action.

Remark 4.1. The negative heat capacity (3.9) implies that the canonical ensemble of asymptotically flat black holes is unstable, and the Euclidean path integral reflects this through a negative mode in the fluctuations around the saddle. Anti-de Sitter (AdS) space acts as a confining box. Hawking and Page (1983) showed that sufficiently large AdS black holes have positive heat capacity and can be in stable equilibrium with their own radiation, and that a first-order phase transition separates thermal AdS space from the black hole phase. This result became central to the interpretation of black holes in the AdS/CFT correspondence (Section 7.8).

¹¹The solution is Ricci-flat, so the Einstein–Hilbert term vanishes and the whole action comes from the boundary term at large radius, after subtraction of the flat-space value.

¹²The observation acquired new importance with the holographic description of the eternal AdS black hole (Section 7.8).

4.3 The stress-energy tensor and the trace anomaly

Particles describe the radiation well only far from the black hole. Near the horizon the right object is the renormalised stress-energy tensor $\langle T_{\mu\nu} \rangle$, which also sources the semiclassical Einstein equations. In two spacetime dimensions, conservation and the trace anomaly determine $\langle T_{\mu\nu} \rangle$ almost completely once the state is specified. Davies et al. (1976) used this to show that, in the Unruh state, the outgoing energy flux at infinity is

$$\langle T_{uu} \rangle|_{\mathcal{H}^+} = \frac{\kappa^2}{48\pi} = \frac{\pi}{12} T_H^2. \quad (4.3)$$

The right-hand side is easy to check. A single right-moving massless boson at temperature T carries the energy flux $\int_0^\infty (d\omega/2\pi) \omega / (e^{\omega/T} - 1) = (T^2/2\pi)(\pi^2/6) = \pi T^2/12$, the energy flux of a one-dimensional black body. For Schwarzschild, $\kappa = 1/4M$ and the flux is $1/(768\pi M^2)$. At the horizon there is instead an ingoing flux of negative energy, $\langle T_{vv} \rangle = -\kappa^2/48\pi$, which is what reduces the mass of the black hole. Christensen and Fulling (1977) extended the analysis to four dimensions by combining the trace anomaly with conservation and regularity conditions.

This derivation establishes what the particle derivations do not: where the energy goes. Near the horizon the dominant effect is the influx of negative energy, and the Hawking flux is not carried by particles that exist, as such, there; the particle interpretation emerges only a few Schwarzschild radii out. Every discussion of backreaction starts from this picture (Section 5.6).

Moving mirrors in two-dimensional flat spacetime provide a closely related model. Fulling and Davies (1976) showed that a perfectly reflecting mirror whose trajectory approaches a null line exponentially, in the manner of Eq. (3.2), produces a thermal flux (Problem 4). The mirror reproduces the kinematics of collapse without any curvature. Carlitz and Willey (1987) later used mirror trajectories to study how correlations in the radiation can restore the purity of the final state, and moving mirrors remain a useful laboratory for questions about information.

4.4 Tunnelling and anomalies

Two further derivations are shorter and less fundamental. In the tunnelling picture the emission rate is set by the imaginary part of the action of a particle crossing the horizon along a classically forbidden path. Srinivasan and Padmanabhan (1999) obtained the temperature from a complex-path analysis of the Hamilton–Jacobi equation. Parikh and Wilczek (2000) used coordinates regular across the horizon and imposed energy conservation on an outgoing s-wave shell of energy ω , so that the mass falls from M to $M - \omega$. They found

$$\Gamma \sim e^{-2\text{Im} S} = \exp\left[-8\pi\omega\left(M - \frac{\omega}{2}\right)\right] = e^{\Delta S_{\text{BH}}}, \quad (4.4)$$

where $\Delta S_{\text{BH}} = 4\pi[(M - \omega)^2 - M^2]$ is the change in the Bekenstein–Hawking entropy. To leading order in ω/M this is the Boltzmann factor $e^{-\omega/T_H}$; the correction is the backreaction of the emitted quantum on the black hole. Writing the rate as $e^{\Delta S_{\text{BH}}}$ suggests a reading in terms of the number of black hole microstates.

Remark 4.2. The tunnelling picture is suggestive but, as usually presented, not a derivation. Done naively in Schwarzschild coordinates, with only the spatial part of the action, it gives twice the Hawking temperature (Problem 3). The correct value is recovered by normalising to the absorption probability, by adding a contribution from the time coordinate, or by using coordinates regular at the horizon. In every version the $i\epsilon$ prescription is a choice of how to continue across the horizon, and that is where regularity of the state enters, unannounced. Even then the method yields a Boltzmann factor for a single quantum, not a spectrum or a state.

Robinson and Wilczek (2005) proposed a derivation based on gravitational anomalies. Near the horizon a field theory on the black hole background reduces to an infinite collection of two-dimensional fields. If the ingoing modes, which classically cannot affect the exterior, are integrated out, the effective theory of the outgoing modes becomes chiral and has a gravitational anomaly. Requiring that general covariance be restored in the full theory fixes the outgoing energy flux to be exactly Eq. (4.3). Iso et al. (2006) extended the argument to charged black holes, where the gauge anomaly fixes the charge flux in the same way. The link with anomaly cancellation is neat, but regularity at the future horizon again enters as a boundary condition, and what comes out is the energy flux, not the spectrum.

4.5 Rigorous results and universality

Hawking's result was placed on a rigorous footing soon after its publication by Wald (1975) and Parker (1975), who showed that the late-time radiation is described by an exactly thermal density matrix (Section 3.5). Fredenhagen and Haag (1990) derived the effect within algebraic quantum field theory. They showed that the late-time flux follows from the assumption that the state is of Hadamard form near the horizon during the collapse, whatever the further details of the initial state.

A related programme asks for the minimal geometrical conditions, following the hint of Remark 3.1. Visser (2003) argued that Hawking radiation is fundamentally kinematical: it requires a Lorentzian geometry with an (apparent) horizon and a well-defined surface gravity, together with a quantum field in a suitable state, but not the Einstein equations, nor a global event horizon, nor the identification of entropy with area. Unruh and Schützhold (2005) showed that the effect survives modifications of the dispersion relation at short wavelengths, under conditions discussed in Lecture 6. Barceló et al. (2011a) identified the essential ingredient as the exponential “peeling” of outgoing null rays, Eq. (3.2), maintained for a sufficiently long time. Objects that approach but never form a horizon can then emit Hawking-like radiation for a while, whereas a horizon that forms without adiabatic peeling need not radiate thermally.

The derivations of Table 4 have two inputs in common: exponential peeling at the rate κ , and a state that looks like the vacuum at short distances across the horizon. Three of them are worth carrying around: Hawking's calculation, because it derives the regularity instead of assuming it; the Unruh–Tolman argument of Section 3.4, because it shows in one line where the temperature comes from; and the Euclidean argument, because it is the fastest. The rest are best read as checks, or as tools for particular questions.

Problems

1. *Euclidean temperature of a static horizon.* For $ds^2 = -f(r)dt^2 + dr^2/f(r) + r^2d\Omega^2$ with a simple zero of f at r_h , show that regularity of the Euclidean section at $r = r_h$ requires $\tau \sim \tau + 4\pi/|f'(r_h)|$, so that $T = |f'(r_h)|/4\pi$. (Hint: set $r - r_h = f'(r_h)\rho^2/4$.) Apply the result to the static patch of de Sitter space, $f = 1 - H^2r^2$, and show that $T = H/2\pi$.
2. *One-dimensional black bodies.* Show that a single right-moving massless boson at temperature T carries the energy flux $\pi T^2/12$, and that a single right-moving massless fermion carries $\pi T^2/24$. (Use $\int_0^\infty x dx/(e^x - 1) = \pi^2/6$ and $\int_0^\infty x dx/(e^x + 1) = \pi^2/12$.) Verify that the first, with $T = \kappa/2\pi$, reproduces Eq. (4.3).
3. *The factor of two in tunnelling.* An outgoing massless s-wave in Schwarzschild coordinates has $S = -\omega t + W(r)$ with $dW/dr = \omega/f$. Passing the pole at $r = 2M$ with the prescription $r - 2M \rightarrow r - 2M - i\epsilon$, show that $\text{Im } W_{\text{out}} = 2\pi M\omega$, and that identifying the emission probability with $e^{-2\text{Im } W_{\text{out}}}$ would give the temperature $2T_H$. With the same prescription

Table 4: Principal derivations of the Hawking effect. All reproduce $T_H = \kappa/2\pi$; they differ in which assumption they make explicit.

Approach	Representative references	Key input	Principal output
Bogoliubov transformation in collapse	Hawking (1975); Wald (1975); Parker (1975)	Vacuum at $\mathcal{I}^-$; geometric optics through the collapsing body	Thermal flux with grey-body factors; exactly thermal density matrix
Detector response and choice of state	Unruh (1976)	State regular on the future horizon	Thermal response of static detectors; relation to the Unruh effect
Euclidean methods	Hartle and Hawking (1976); Gibbons and Hawking (1977a)	Regularity of the Euclidean section at the horizon	Temperature from periodicity; entropy $A/4$ from the on-shell action
Stress tensor and trace anomaly	Davies et al. (1976); Christensen and Fulling (1977)	Conformal anomaly, conservation, regularity	Energy flux at infinity; negative energy influx at the horizon
Tunnelling	Srinivasan and Padmanabhan (1999); Parikh and Wilczek (2000)	WKB amplitude across the horizon; energy conservation	Boltzmann factor, with corrections from backreaction
Gravitational anomalies	Robinson and Wilczek (2005); Iso et al. (2006)	General covariance of the near-horizon effective theory	Two-dimensional black-body flux at T_H
Algebraic quantum field theory	Kay and Wald (1991); Fredenhagen and Haag (1990)	Hadamard condition on the state	Uniqueness and thermality (KMS property) of the stationary state

an ingoing wave, $dW/dr = -\omega/f$, has $\text{Im } W_{\text{in}} = -2\pi M\omega$. Show that the ratio of emission and absorption probabilities, $e^{-2\text{Im } W_{\text{out}}}/e^{-2\text{Im } W_{\text{in}}}$, is the Boltzmann factor at T_H .

4. *A thermal mirror.* For a mirror in two-dimensional Minkowski spacetime, an incoming ray at advanced time v is reflected into the outgoing ray at retarded time u with $v = p(u)$, and in the in-vacuum the reflected flux is $\langle T_{uu} \rangle = (1/24\pi) \left[\frac{3}{2}(p''/p')^2 - p'''/p' \right]$. Show that the trajectory $p(u) = v_0 - C e^{-\kappa u}$, the analogue of Eq. (3.2), gives exactly $\kappa^2/48\pi$ at all u . Show also that a mirror moving with constant velocity, for which p is linear in u , radiates nothing.
5. *Which state is unique?* Identify the state of Table 3 to which the theorem of Kay and Wald (1991) applies on the extended Schwarzschild spacetime. Explain why the Unruh state, which is stationary and regular on the future horizon, does not contradict the uniqueness, and why its failure on the past horizon is harmless for a black hole formed by collapse.

Notes and further reading

The short paper of Gibbons and Hawking (1977a) is the one to read first: the Euclidean argument and the entropy calculation take a few pages and have hardly been improved on. Kay and Wald (1991) is long and mathematically demanding, but the statements of its main theorems can be understood without the proofs, and the reader who wants to know exactly what “thermal” means for the states of Table 3 should study at least those. Birrell and Davies (1982) treat the stress-energy tensor in two dimensions, including the moving-mirror formula of Problem 4, in more detail than we have done; the original computation of Davies et al. (1976) is brief and still readable.

For the question of what is essential, Visser (2003) is a clear statement of the kinematical view and Barceló et al. (2011a) sharpens it into precise conditions on the peeling of null rays. Both are good preparation for the discussion of analogue systems in Lecture 8.

Lecture 5 Evaporation

A black hole of one solar mass radiates some 10^{-28} W and, left alone in empty space, would need about 10^{67} years to evaporate. A black hole of 5×10^{14} g, roughly the mass of a block of rock 600 m on a side, has a temperature of about 20 MeV; if one formed in the early universe, it is finishing its evaporation now. Both statements follow from Eq. (3.6); the second also needs some particle physics.

5.1 Greybody factors

The greybody factor is the answer to a one-dimensional scattering problem. For a massless field of spin $s = 0, 1, 2$ on the Schwarzschild background, separation of variables leaves a radial function ψ that obeys a wave equation in the tortoise coordinate,

$$\frac{d^2\psi}{dr_*^2} + [\omega^2 - V_{s\ell}(r)] \psi = 0, \quad V_{s\ell}(r) = \left(1 - \frac{2M}{r}\right) \left[\frac{\ell(\ell+1)}{r^2} + \frac{2M(1-s^2)}{r^3} \right]. \quad (5.1)$$

For $s = 2$ this is the Regge–Wheeler equation for axial gravitational perturbations. Fields of half-integer spin obey analogous equations, and the Teukolsky formalism treats all spins on the Kerr background at once (Teukolsky and Press, 1974). The potential vanishes at the horizon ($r_* \rightarrow -\infty$) and at infinity, with a maximum near the photon sphere $r = 3M$. The greybody factor $\Gamma_{s\ell}(\omega)$ is the probability of transmission through this barrier. By time reversal it is also the probability that a wave sent in from infinity is absorbed, which is how it is computed in practice.

Two limits fix the shape of Γ . They are best stated through the absorption cross-section per polarisation,

$$\sigma(\omega) = \frac{\pi}{\omega^2} \sum_{\ell \geq s} (2\ell + 1) \Gamma_{s\ell}(\omega).$$

At high frequencies, $M\omega \gg 1$, geometrical optics applies. A partial wave of angular momentum ℓ has impact parameter $b \approx (\ell + \frac{1}{2})/\omega$, and it passes over the barrier when b is below the critical value $3\sqrt{3}M$ set by the photon sphere. Then $\Gamma_{s\ell} \approx 1$ for $\ell \lesssim 3\sqrt{3}M\omega$ and $\Gamma_{s\ell} \approx 0$ above, the sum is $(3\sqrt{3}M\omega)^2$, and $\sigma \rightarrow 27\pi M^2$, the geometric capture cross-section of the photon sphere. At low frequencies, $M\omega \ll 1$, the barrier is nearly opaque, the more so the higher the spin and the angular momentum. For the s-wave of a massless scalar field $\Gamma_{00} \approx 16M^2\omega^2$, and the low-frequency cross-section is $16\pi M^2$, which is exactly the area of the horizon (Das et al., 1997).¹³ Fields of higher spin start at $\ell = s$, and their lowest partial wave is transmitted with a higher power of ω . Their emission is therefore suppressed relative to a black body, and the spectrum peaks at somewhat higher energies than a Planck spectrum at the same temperature.

5.2 Luminosity, mass loss, and lifetime

The scaling of the luminosity is known in advance: an area of order M^2 radiating at a temperature of order $1/M$ gives $L \propto M^{-2}$, and since M is the only scale in the problem this is exact. A first guess treats the black hole as a black body at temperature T_H whose emitting area is the

¹³Das et al. (1997) showed that this equality holds for every spherically symmetric black hole, not only for Schwarzschild.

high-frequency cross-section $27\pi M^2$. For photons, with the Stefan–Boltzmann constant $\pi^2/60$ in natural units, this gives

$$L_\gamma \approx \frac{\pi^2}{60} \times 27\pi M^2 \times \left(\frac{1}{8\pi M} \right)^4 = \frac{27}{245760\pi} \frac{1}{M^2} \approx 3.5 \times 10^{-5} \frac{1}{M^2}. \quad (5.2)$$

The coefficient was first computed by Page (1976a) from the greybody factors of all massless fields on the Schwarzschild background. He assumed the particle content then believed to be massless: two species of two-component neutrinos with their antiparticles, the photon, and the graviton. Approximately 81% of the power is then carried by neutrinos, 17% by photons and 2% by gravitons, and the total mass-loss rate is

$$\frac{dM}{dt} = -\frac{\alpha}{M^2}, \quad \alpha \simeq 2.0 \times 10^{-4}. \quad (5.3)$$

The photon share, about $3.4 \times 10^{-5}/M^2$, is close to (5.2). Students sometimes take this as evidence that the estimate is sound; the agreement is fortuitous. The same estimate overestimates the graviton power, which Page found to be about $3.8 \times 10^{-6}/M^2$, by nearly an order of magnitude. Applied to Page’s four neutrino states, each of a single helicity and with the fermionic factor $7/8$, it gives about $6 \times 10^{-5}/M^2$, well below Page’s $1.6 \times 10^{-4}/M^2$. The greybody factors depend strongly on spin, and no single effective area describes all species.¹⁴

Equation (5.3) is readily integrated. If α were constant, M^3 would decrease linearly in time, and

$$M(t) = M_0 \left(1 - \frac{t}{\tau} \right)^{1/3}, \quad \tau = \frac{M_0^3}{3\alpha}. \quad (5.4)$$

The black hole evaporates completely in a finite time τ . Most of that time is spent near the initial mass: the mass reaches $M_0/2$ only at $t = 7\tau/8$, and the end is a runaway, with the luminosity diverging as $(\tau - t)^{-2/3}$. Restoring units, $\tau = G^2 M_0^3 / (3a\hbar c^4)$, and Page’s value of α gives $\tau \simeq 8.7 \times 10^{-27} (M_0/1 \text{ g})^3 \text{ s}$.

For a solar-mass black hole the temperature is about $5 \times 10^{-12} \text{ eV}$, far below the neutrino masses implied by oscillation experiments. Only photons and gravitons are emitted in appreciable numbers, the effective α drops to about 3.7×10^{-5} , and the lifetime is of the order of 10^{67} years.¹⁵ (Today such a black hole in fact absorbs far more from the microwave background than it emits; see Problem 4.) At smaller masses α grows, as new species become available (Section 5.4).

5.3 Rotation and charge

Page (1976b) showed that a Kerr black hole loses angular momentum considerably faster than it loses mass. The reason is visible in Eq. (3.6): the factor $\omega - m\Omega_H$ favours quanta with large m , which carry away angular momentum efficiently, and the effect is stronger for fields of higher spin. The final spin therefore depends on what is emitted. Chambers et al. (1997) found that a black hole emitting only scalar quanta does not spin down to zero; its dimensionless spin a/M tends instead to about 0.555.

Charge is shed efficiently as well. A charged black hole preferentially emits particles with the same sign of charge, and, when the electric field near the horizon exceeds the Schwinger critical value, it also discharges by vacuum pair production (Damour and Ruffini, 1976; Gibbons,

¹⁴With three flavours of light neutrinos instead of Page’s two, the same greybody factors give $\alpha \simeq 2.8 \times 10^{-4}$. The choice of $27\pi M^2$ as emitting area is itself rough; see Problem 2.

¹⁵Oscillations fix only mass differences, so the lightest neutrino could be nearly massless. If it were lighter than about 10^{-11} eV it would still be emitted, α would roughly triple, and the lifetime would fall by the same factor.

1975).¹⁶ The emission of charged leptons from a non-rotating black hole was computed by Page (1977). Black holes light enough to evaporate on cosmological time scales should therefore be nearly neutral, and should have lost most of their spin long before the end.

5.4 Massive particles and secondary emission

A species of mass m is emitted in appreciable numbers only when $T_H \gtrsim m$. As a black hole evaporates, its temperature climbs through the thresholds of the Standard Model, and the number of degrees of freedom contributing to α increases. Electrons appear near $M \approx 2 \times 10^{16}$ g, where $T_H \approx m_e$. Once T_H exceeds the QCD scale, about 0.2 GeV (reached near 5×10^{13} g), quarks and gluons are emitted as elementary particles. They fragment and hadronise at distances of order a fermi, which is large compared with the black hole itself (Problem 5).

MacGibbon and Webber (1990) showed that secondary particles, in particular photons from neutral pion decay (centred near $m_{\pi^0}/2 \approx 68$ MeV), dominate the photon spectrum at high temperatures, and MacGibbon (1991) integrated the emission over the lifetime. With all Standard Model species active, α reaches about 4×10^{-3} , some twenty times Page's value, and a black hole radiates its last 10^9 g or so in about a second (Table 2). With the full Standard Model, a black hole of initial mass approximately 5×10^{14} g has a lifetime equal to the present age of the universe (Carr et al., 2010; MacGibbon et al., 2008); Page's value of α alone would give a somewhat smaller mass (Problem 1). The public code `BLACKHAWK` (Arbey and Auffinger, 2019, 2021) computes primary and secondary spectra for arbitrary distributions of masses and spins and is now standard in the analyses of Lecture 9.

Whether the emitted particles interact enough to modify the spectrum has been debated. Heckler (1997) argued that bremsstrahlung and pair production would build a photosphere, and a corresponding QCD "chromosphere", around sufficiently hot black holes, degrading the spectrum to lower energies. MacGibbon et al. (2008) replied that causal and kinematical considerations prevent such a photosphere from forming, so that the particles stream freely. Current analyses adopt the latter view, and so do we.

5.5 A sparse flux

Hawking radiation differs from the glow of a hot body in one respect that is easy to miss: it is extremely sparse. Take the photon luminosity (5.2) and a mean photon energy of about $2.7 T_H \approx 0.11/M$. The emission rate is then of order $3 \times 10^{-4}/M$, so successive photons are separated on average by a time of order $3 \times 10^3 M$. The period of a typical photon, $2\pi/\langle\omega\rangle$, is only about $60M$. Successive quanta are therefore separated by some fifty wave periods.

The reason is geometrical. For any black body of area A at temperature T , the number of photons emitted per period scales as AT^2 , the area measured in units of the thermal wavelength squared. For a lamp filament AT^2 is enormous. For a black hole $A \sim M^2$ and $T \sim 1/M$, so AT^2 is a pure number, and it happens to be small (Problem 3). Gray et al. (2016) made this observation precise, and it had been emphasised earlier by Page (2005). The Hawking flux is better pictured as a slow sequence of individual quanta than as a continuous thermal fluid.

5.6 Backreaction and the end point

So far the black hole has been a fixed background. Its evolution is usually described in the semiclassical approximation, in which the metric is classical and sourced by the expectation value of the renormalised stress-energy tensor, $G_{\mu\nu} = 8\pi\langle T_{\mu\nu}\rangle$. Is the approximation consistent?

¹⁶For electrons the critical field is $E_c = m_e^2 c^3 / e\hbar \approx 1.3 \times 10^{18}$ V m⁻¹.

The test is adiabaticity: the surface gravity should change little over the time κ^{-1} on which the emission is established. With $\kappa = 1/4M$ and Eq. (5.3),

$$\frac{1}{\kappa^2} \left| \frac{d\kappa}{dt} \right| = 4 \left| \frac{dM}{dt} \right| = \frac{4\alpha}{M^2} \ll 1. \quad (5.5)$$

For a solar-mass black hole the left-hand side is below 10^{-79} . It reaches unity only when M is about a tenth of the Planck mass or less, where nothing here can be trusted anyway. Bardeen (1981) argued that, with backreaction included, evaporation is quasi-static and the emission remains thermal at the instantaneous temperature until the Planck scale is approached. York (1983) constructed a quantum-corrected geometry describing the dynamical origin of the radiation near the horizon.

Two-dimensional dilaton gravity provides solvable models with backreaction. Callan et al. (1992) introduced a model in which formation and evaporation can be followed with one-loop effects included, and Russo et al. (1992) found a modification in which the end point of evaporation is an exact solution. These models confirm that semiclassical evaporation is controlled down to the scale where quantum gravity becomes strong. What happens at that scale in four dimensions they cannot tell us.

The end point is therefore unknown. The possibilities discussed include complete evaporation, leaving only radiation; a stable or long-lived Planck-scale remnant (Aharonov et al., 1987); and a transition to some other object whose description requires quantum gravity. A remnant would have to store arbitrarily much information in a Planck-scale region, with an unbounded number of internal states; the arguments for and against have been reviewed by Chen et al. (2015). Dvali et al. (2020) proposed that the information stored in a black hole exerts a “memory burden” that strongly suppresses the emission once a substantial fraction of the initial mass has been lost. The proposal lies outside the semiclassical framework and remains speculative. If correct, it would substantially modify the constraints on primordial black holes discussed in Lecture 9 (Alexandre et al., 2024; Thoss et al., 2024).

Remark 5.1. Where does the radiation originate? The pair picture of Section 1.3 suggests a thin layer at the horizon. Taken literally, the picture is wrong. The wavelength of a typical quantum, $2\pi/(2.7 T_H) \approx 60M$, is much larger than the black hole, and the stress-tensor analyses of Section 4.3 show the outgoing flux building up well outside the horizon. Giddings (2016) argued, from the Stefan–Boltzmann law and the effective radius of emission, that the radiation originates in a “quantum atmosphere” extending a distance of order the horizon radius beyond the horizon. Dey et al. (2017) reached a broadly compatible conclusion from the renormalised stress-energy tensor and the correlations between quanta and partners. The location matters both for the trans-Planckian problem of Lecture 6 and for proposals in which the near-horizon region is modified (Lecture 7).

Problems

1. Restore units in Eq. (5.4) and show that $\alpha = 2.0 \times 10^{-4}$ gives $\tau \simeq 8.7 \times 10^{-27} (M_0/1 \text{ g})^3 \text{ s}$. Find the initial mass that evaporates in 13.8 Gyr with this constant α (about $3.7 \times 10^{14} \text{ g}$). Defining an effective constant $\bar{\alpha}$ by $\tau = M_0^3/3\bar{\alpha}$, show that $M_0 = 5 \times 10^{14} \text{ g}$ requires $\bar{\alpha} \approx 5 \times 10^{-4}$, and say which particles account for the difference.
2. Using Eq. (3.6), show that the power emitted in one polarisation state of a massless boson is $(1/2\pi^2) \int_0^\infty \sigma(\omega) \omega^3 d\omega / (e^{\omega/T_H} - 1)$, with σ the absorption cross-section of Section 5.1. Deduce that taking $\sigma = 27\pi M^2$ at all frequencies gives four times (5.2) for photons, and explain the factor of four by comparing the cross-section and the surface area of a black sphere. What does the ratio of Page’s photon power to this value tell you about the low-frequency greybody factors?

3. For a black body of area A at temperature T , the photon number flux per unit area is $\zeta(3)T^3/2\pi^2$ and the mean photon energy is $\pi^4 T/30\zeta(3) \approx 2.7T$. Show that the mean time between emitted photons, divided by the period $2\pi/\langle\omega\rangle$, is approximately $7/(AT^2)$. Evaluate it for a black hole with $A = 27\pi M^2$ and $T = T_H$ (about 50), and for a 1 cm^2 surface at 300 K (about 4×10^{-6}).
4. A solar-mass black hole sits in the cosmic microwave background at $T_{\text{CMB}} = 2.725\text{ K}$. Taking the absorption cross-section to be $27\pi M^2$ (justify this for the relevant frequencies) and the emitted power to be $3.7 \times 10^{-5}/M^2$, show that the absorbed power exceeds the emitted power by a factor of order 10^{30} .
5. A black hole of 10^{13} g has $T_H \approx 1\text{ GeV}$. Compute its Schwarzschild radius (about 0.015 fm) and the wavelength $2\pi/\langle\omega\rangle$ of a typical emitted quantum, with $\langle\omega\rangle \approx 2.7 T_H$ (about 0.4 fm). Use these numbers, and the size of a hadron, to explain why such a black hole emits quarks and gluons rather than pions, and why hadronisation takes place far outside the horizon.

Notes and further reading

The numbers in this lecture rest on the papers of Page (1976a,b), which remain the place to see how greybody factors are computed and summed; the first should be consulted by anyone who wants to check a number given here. For the Standard Model era of the subject, MacGibbon and Webber (1990) explains how quark and gluon jets enter, and Carr et al. (2010) collects the lifetime and the emitted spectra in a form convenient for cosmological applications. Auffinger (2023) is the most useful recent review for a student who wants to compute spectra, and it pairs naturally with the BLACKHAWK manual (Arbey and Auffinger, 2019).

On the end point there is far less to read that is settled. Chen et al. (2015) give a fair account of the remnant debate, and Page (2005) discusses the late stages, including the sparseness of the flux, from the point of view of the information problem we take up in Lecture 7.

Lecture 6 The trans-Planckian problem

Hawking's calculation treats the field as free and the geometry as fixed at every wavelength, however short. For almost all of the radiation, that assumption is applied far below the Planck length.

6.1 Statement of the problem

Read the exponential relation (3.2) backwards in time. A quantum of frequency $\omega \sim T_H$ that reaches $\mathcal{I}^+$ at retarded time u corresponds, near the horizon and in the frame of a freely falling observer, to a mode whose frequency grows as $\omega e^{\kappa u}$.¹⁷ The frequency reaches the Planck mass after a time

$$u_P \sim \frac{1}{\kappa} \ln \frac{m_P}{T_H} = 4M \ln(8\pi M), \quad (6.1)$$

where, in the last expression, M is measured in Planck units. For a solar-mass black hole, $4M \approx 2 \times 10^{-5}\text{ s}$ and $\ln(8\pi M/m_P) \approx 90$, so u_P is a couple of milliseconds. The logarithm makes this short for every black hole: for one of $5 \times 10^{14}\text{ g}$ it is a few times 10^{-22} s (Problem 1). Essentially all the radiation therefore appears to come from modes whose wavelengths near the horizon were far below the Planck length, where a free field on a fixed background is not expected

¹⁷Frequency is not a Lorentz invariant; the statement refers to a fixed family of free-fall frames, for example observers dropped from rest far away at successive times (see Section 6.4).

to be a good description. 't Hooft (1985) emphasised the difficulty, arguing that gravitational interactions near the horizon become strong, and Jacobson (1991) made it a sharp problem.

The logical status of the problem is often garbled. It is not an inconsistency in Hawking's derivation, which is correct for a free field on a fixed background, but the question of whether the result survives different short-distance physics. That cannot yet be settled in quantum gravity, but one can change the short-distance physics by hand in a model.

6.2 Dispersive models

Unruh (1995) did this in the acoustic analogue described in Lecture 8.¹⁸ In a fluid flowing with velocity $v(x)$, a sound wave of laboratory frequency ω and wavenumber k obeys a dispersion relation of the form

$$(\omega - vk)^2 = F(k)^2, \quad F(k) \simeq c_s |k| \quad \text{for } |k| \ll k_d, \quad (6.2)$$

where $\omega - vk$ is the frequency in the frame comoving with the fluid, ω is conserved in a stationary flow, and k_d marks the departure from linearity. In the gravitational analogy k_d plays the part of the Planck scale, and with linear dispersion the fluid reproduces the problem exactly (Problem 2). Real fluids, however, have subluminal dispersion: $F(k)$ grows more slowly than $c_s |k|$ at large $|k|$. Unruh solved the wave equation numerically and found the thermal spectrum at $T_H = \kappa/2\pi$ to high accuracy, although no mode ever attains an arbitrarily short wavelength.

With linear dispersion, an outgoing mode traced backwards in time hugs the horizon and is blueshifted without limit. With subluminal dispersion, short waves travel more slowly than long ones, and traced backwards the mode is eventually swept away from the horizon by the flow. Its ancestor is not a trans-Planckian mode lingering at the horizon but an ingoing short-wavelength mode that is converted near the horizon into an outgoing long-wavelength one. This mode conversion replaces the infinite blueshift. For superluminal dispersion, such as the Bogoliubov relation of a Bose–Einstein condensate, $F(k)^2 = c_s^2 k^2 + (k^2/2m)^2$, the ancestral modes come instead from inside the horizon, but the conclusion is the same.

Jacobson (1993) imposed a short-distance cutoff on the field modes, Brout et al. (1995) analysed the problem in their review, and Corley and Jacobson (1996) computed the spectrum for both kinds of dispersion: it is thermal provided that $k_d c_s \gg \kappa$.

6.3 Conditions for universality

A common misreading is that dispersion removes the high frequencies from the problem. It does not: the short-wavelength modes are still there, as the ingoing ancestors of the Hawking quanta, and their state matters. Unruh and Schützhold (2005) showed that, for a broad class of dispersion relations, the effect is universal under two conditions. The evolution of the relevant modes near the horizon must be adiabatic, which holds when the dispersion scale greatly exceeds κ . And the ingoing short-wavelength modes must be in their ground state.

The first condition is kinematical and easily met. For a solar-mass black hole κ is about 3×10^{-39} in Planck units, so Planck-scale dispersion is adiabatic by an enormous margin. The second is different in kind: in gravity it is an assumption about the ultraviolet completion of the theory. The problem is therefore not so much resolved as relocated: the prediction is insensitive to short-distance physics provided that those degrees of freedom are unexcited. This is the content of the Hadamard condition of Fredenhagen and Haag (1990) (Section 4.5).

Remark 6.1. Dispersion also produces effects with no counterpart in the linear theory. With superluminal dispersion, in a flow with both a black hole horizon and an inner, white hole,

¹⁸The analogy itself was introduced by Unruh (1981); the 1995 paper added the dispersion.

horizon, modes bounce between the two and are amplified at each passage. The resulting exponential instability is what Corley and Jacobson (1999) called a black hole laser; it has been reported in a Bose–Einstein condensate (Steinhauer, 2014) (Section 8.4).

6.4 What remains open

It helps to separate what has been shown from what has been assumed. For linear field theories with modified dispersion on stationary backgrounds, it is established that adiabaticity and a ground state for the ingoing short-wavelength modes are enough to give a thermal flux at $\kappa/2\pi$, with corrections that vanish as $\kappa/k_d c_s \rightarrow 0$. The experiments of Lecture 8 probe the same mechanism in real media.

Carrying these results over to gravity requires two assumptions. One is that the ultraviolet behaves like a dispersive medium. That needs a preferred frame, hence a violation of local Lorentz invariance at high energies, which nothing obliges quantum gravity to supply. The other is the ground-state condition, which no model can derive without knowing the high-energy theory. There is also a gap: the models are free field theories, whereas the concern of 't Hooft (1985) was the gravitational interaction between ingoing and outgoing quanta.

A different argument avoids the preferred frame. The relevant modes need never be Planckian in the frame of an infalling observer at the moment the Hawking quanta separate from the vacuum, because this happens only once their wavelength is comparable to the size of the black hole (Jacobson, 2005; Polchinski, 2017). This fits the extended emission region of Remark 5.1. For gravity we find it the more convincing argument, but it concerns where the physics happens; it is not a calculation in quantum gravity, which a complete resolution presumably requires. Still, no modification of short-distance physics that preserves both conditions has been found to eliminate the effect, and that is the real basis for confidence in the prediction.

Problems

1. Derive Eq. (6.1) by setting $T_H e^{\kappa u}$ equal to the Planck mass. Evaluate u_P for a solar-mass black hole (about 2 ms) and for a black hole of 5×10^{14} g (about 2×10^{-22} s), and compare the second with that black hole's lifetime.
2. Model the flow near a sonic horizon by a constant c_s and $v(x) = -c_s + \kappa x$, so that the horizon is at $x = 0$. With linear dispersion, show that an outgoing ray outside the horizon obeys $x(t) = x_0 e^{\kappa t}$, and, using conservation of ω on the branch $\omega - vk = c_s k$, that its wavenumber is $k(t) = (\omega/\kappa x_0) e^{-\kappa t}$. Find the time $t_d < 0$ at which k reached k_d and show that it has the logarithmic form of Eq. (6.1).
3. For the Bogoliubov relation $F(k)^2 = c_s^2 k^2 + (k^2/2m)^2$ (with $\hbar = 1$), show that the group velocity dF/dk exceeds c_s for every $k > 0$. Identify the dispersion scale as the wavenumber at which the two terms are equal, $k_d = 2mc_s$, and express the condition of Corley and Jacobson (1996) as a condition on mc_s^2 and κ . Explain why superluminal propagation allows the ancestral modes to come from inside the horizon.
4. A colleague argues that, because the Hawking quanta of a solar-mass black hole descend from trans-Planckian modes, the prediction $T_H = \kappa/2\pi$ cannot be trusted. Write a half-page reply stating what the dispersive models establish, which of the two conditions of Section 6.3 carries the weight in gravity and why, and one piece of physics that the models leave out.

Notes and further reading

Jacobson (1991) is short and states the problem clearly; read it first. Unruh (1995) is the paper that changed the discussion, and Corley and Jacobson (1996) is the analytical treatment to work through with a pencil. The conditions for universality are set out in Unruh and Schützhold (2005). The lecture notes of Jacobson (2005) give a balanced account of the whole question, including the argument from the infalling frame.

For a sceptical view of the standard derivations, which takes the trans-Planckian problem more seriously than most authors do, see Helfer (2003). We recommend it precisely because it disagrees with the consensus presented here.

Lecture 7 Black hole entropy and information

A black hole of one solar mass has a Bekenstein–Hawking entropy $S_{\text{BH}} = 4\pi M^2 \approx 10^{77}$. The Sun has a thermal entropy of roughly 10^{58} . Collapse to a black hole would therefore raise the entropy of a star by some nineteen orders of magnitude, and if S_{BH} is a statistical entropy in the ordinary sense, the black hole has of order $e^{10^{77}}$ microstates. Hawking’s calculation fixed the coefficient in $S = A/4$ and left two questions behind. The first is what those states are. The second, which Hawking himself posed a year later, is what becomes of the information carried by matter that falls into a black hole which then evaporates. The second has occupied quantum gravity for five decades, and it takes up most of this lecture.

7.1 What the entropy counts

The first precise count of black hole microstates came from string theory. Strominger and Vafa (1996) considered a class of extremal black holes in five dimensions carrying three charges. At weak string coupling the same charges are carried by bound states of Q_1 D1-branes and Q_5 D5-branes with N units of momentum along a common circle, and the number of supersymmetric bound states grows, for large charges, as $e^{2\pi\sqrt{Q_1 Q_5 N}}$. At strong coupling the same system is a black hole with $A/4 = 2\pi\sqrt{Q_1 Q_5 N}$, numerical coefficient included. The match is possible because supersymmetry protects the number of states as the coupling is varied.

The programme was soon extended to near-extremal black holes, which have a small Hawking temperature. Callan and Maldacena (1996) showed that excited D-brane configurations emit closed strings with the qualitative features of Hawking radiation, Das and Mathur (1996) that the rates agree with the semiclassical ones, and Maldacena and Strominger (1997) that the agreement extends to the greybody factors. In these settings Hawking emission is the decay of excited open strings on the branes into closed strings, a manifestly unitary process. The results are restricted to black holes close to extremality, mostly in supersymmetric theories; none applies directly to Schwarzschild.

In loop quantum gravity, Rovelli (1996) and Ashtekar et al. (1998) counted quantum geometry states of the horizon and obtained an entropy proportional to A , with a coefficient that depends on the Barbero–Immirzi parameter. Fixing that parameter by demanding $A/4$ calibrates the theory rather than predicting the coefficient, although the value so fixed then serves for the whole class of non-rotating horizons treated by Ashtekar et al. (1998).

A third idea needs no theory of quantum gravity at all. Bombelli et al. (1986) and Srednicki (1993) showed that the vacuum entanglement entropy of a quantum field across a surface of area A behaves as $\kappa A/\epsilon^2$, with ϵ an ultraviolet cutoff and κ a non-universal number. With ϵ of order the Planck length this is of order S_{BH} , which suggests that part of the black hole entropy is entanglement between interior and exterior; the dependence on κ shows that this is not yet a

derivation. The cleaner statement, reviewed by Solodukhin (2011), is that the same divergences renormalise Newton’s constant, so that the entanglement of the fields and the bare term $A/4G_{\text{bare}}$ combine into $A/4G$. We return to entanglement and area in Section 7.8. With higher-curvature terms in the action, finally, $A/4$ is replaced by the Noether charge entropy of Wald (1993).

7.2 Hawking’s argument

The argument of Hawking (1976) takes a few lines given the state (3.8) of Lecture 3. Suppose that a black hole forms from matter in a pure quantum state. Each outgoing Hawking quantum is entangled with a partner inside, and the radiation on its own is in an exactly thermal mixed state. While the black hole exists, the joint state of radiation and interior is pure. If the black hole evaporates completely, however, the partners disappear with it and the final state consists of thermal radiation alone. The evolution would then take a pure state to a mixed one, which no unitary evolution can do. Hawking concluded that the S -matrix of quantum gravity must be replaced by a “superscattering” operator mapping density matrices to density matrices, with a fundamental loss of information.¹⁹

It helps to state the tension as an incompatibility between three principles, each well supported on its own. Principle (i) is the validity of semiclassical physics and of the equivalence principle near the horizon of a large black hole, where the curvature is of order $1/M^2$ and nothing locally distinguishes the horizon from any other null surface. Principle (ii) is unitarity, as seen by observers who stay outside. Principle (iii) is the locality of effective field theory, which forbids information from being in two places at once or from travelling outside the light cone. Hawking’s argument uses (i) and (iii) to derive the violation of (ii). Every proposed resolution can be classified by which of the three it modifies, and the reader should ask this of each proposal.

A student who first meets the argument often locates the difficulty at the very end of the evaporation, when the black hole reaches the Planck scale and semiclassical physics fails anyway. Page’s analysis shows that this is wrong: the conflict appears when the black hole is still large.

7.3 Page’s theorem and the Page curve

Page (1993a) began with a result about random pure states. Choose a pure state at random (with the unitarily invariant measure) on a Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$ of dimensions $m \leq n$. The average entanglement entropy of the smaller subsystem is

$$\langle S_A \rangle \simeq \ln m - \frac{m}{2n}, \quad (7.1)$$

which lies within half a unit of its maximum value $\ln m$.²⁰ For ten qubits inside a typical pure state of 110 qubits, $m = 2^{10}$ and $n = 2^{100}$, and the entropy falls short of $10 \ln 2$ by 2^{-91} . A small subsystem of a typical pure state is very nearly maximally mixed and carries almost no information about the whole. Problem 1 derives a weaker bound of the same type.

Page (1993b) then applied the theorem to evaporation. Suppose that evaporation is unitary and that the joint state of black hole and radiation behaves like a typical pure state. Early on, the radiation is the smaller subsystem and its entropy grows, as in Hawking’s calculation. Once more than half of the degrees of freedom have been emitted, the black hole is the smaller subsystem, and the entropy of the radiation must equal that of the black hole, which is decreasing. The entropy of the radiation therefore rises and then falls back to zero. This is the Page curve; its maximum is at the Page time. The information problem can now be put sharply: a unitary theory must reproduce the Page curve, and the semiclassical calculation does not.

¹⁹Hawking denoted this operator by a dollar sign, $\$$, and it is still sometimes called the dollar matrix.

²⁰Page conjectured the exact result, $\langle S_A \rangle = \sum_{k=n+1}^{mn} 1/k - (m-1)/(2n)$, from which Eq. (7.1) follows for $1 \ll m \leq n$; proofs appeared within three years.

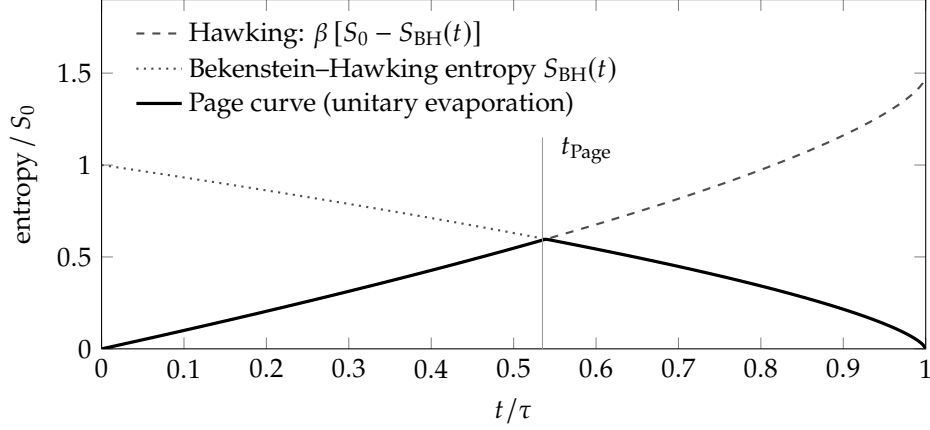


Figure 2: Entropy of the Hawking radiation against time for a black hole of lifetime τ and initial entropy S_0 , from Eq. (7.2) with $\beta = 1.48$. In Hawking's calculation (dashed) the entropy of the radiation grows until the end of the evaporation. Unitarity requires the fine-grained entropy to follow the Page curve (solid), which turns over at the Page time and returns to zero.

The curve is easy to estimate. Since $S_{\text{BH}} \propto M^2$, Eq. (5.4) gives $S_{\text{BH}}(t) = S_0(1 - t/\tau)^{2/3}$. Emission into empty space is irreversible, so the coarse-grained entropy of the emitted radiation exceeds the decrease of S_{BH} by a factor $\beta > 1$, which Page (2013) estimated to be approximately 1.5 for a black hole emitting photons and gravitons. In Hawking's calculation the entropy of the radiation is this coarse-grained entropy, $S_{\text{Hawking}}(t) = \beta[S_0 - S_{\text{BH}}(t)]$, whereas unitarity bounds the fine-grained entropy of the radiation by that of the black hole. The Page curve is therefore approximately

$$S_{\text{Page}}(t) \simeq \min \{ \beta [S_0 - S_{\text{BH}}(t)], S_{\text{BH}}(t) \}. \quad (7.2)$$

The two branches cross when $S_{\text{BH}}/S_0 = \beta/(1 + \beta) \approx 0.6$, that is, at $t_{\text{Page}} \approx 0.54 \tau$. At that moment the black hole has lost only about a quarter of its initial mass. For a solar-mass black hole this is some 10^{67} years after formation, and the horizon is still kilometres across. The three curves are shown in Fig. 2.

Remark 7.1. The Page curve is not derived from any dynamics: it is what unitarity plus typicality demands, and it concerns the fine-grained (von Neumann) entropy of the radiation. The coarse-grained entropy keeps growing until the end in any account. The descending branch does not mean that late radiation looks less thermal. Each quantum looks as thermal as before; correlations among many quanta purify the whole.

Page's analysis also shows how well information can be hidden. Before the Page time essentially nothing about the initial state can be extracted from the radiation, even if the evaporation is unitary, and afterwards the information sits in correlations among very many quanta. No experiment on a few Hawking quanta can test unitarity.

7.4 Old black holes as mirrors

Hayden and Preskill (2007) added a twist. Model the black hole dynamics as a fast random unitary transformation, and take a black hole past its Page time, maximally entangled with the earlier radiation. Throw in a diary of k qubits. An observer who holds the early radiation, and knows the dynamics, can reconstruct the diary from only slightly more than k qubits of subsequent radiation, after a time of order the scrambling time

$$t_* \sim \frac{1}{2\pi T_{\text{H}}} \ln S_{\text{BH}} = 4M \ln S_{\text{BH}}, \quad (7.3)$$

which is of order $M \ln M$. The old black hole behaves as an information mirror. The scrambling time is short: for a solar-mass black hole it is a few milliseconds (Problem 4).

7.5 Why small corrections do not help

The natural first hope is that small corrections to Hawking's calculation accumulate over the 10^{77} or so emissions and purify the radiation. Mathur (2009) showed that this hope is unfounded, by an argument short enough to give in full.

Treat the evaporation as a sequence of steps, at each of which a new pair is created near the horizon: an outgoing quantum b , which joins the radiation, and its partner c , which remains inside. Model each by a qubit, so that at leading order the pair is in a maximally entangled state, the qubit version of Eq. (3.8): then $S(bc) = 0$ and $S(b) = \ln 2$. Let R be the radiation emitted earlier. The entropy of the radiation changes from $S(R)$ to $S(Rb)$, and the Page curve requires $S(Rb) < S(R)$ after the Page time.

The tool is strong subadditivity, which holds for any state of any three systems X, Y, Z :²¹

$$S(XY) + S(YZ) \geq S(Y) + S(XYZ). \quad (7.4)$$

Take $X = R, Y = b, Z = c$, and allow the new pair to deviate from the Hawking state, by assuming only that $S(bc) \leq \epsilon$. The Araki–Lieb inequality gives $S(Rbc) \geq S(R) - S(bc)$, and substituting into Eq. (7.4) yields

$$S(Rb) \geq S(R) + S(b) - 2S(bc) \geq S(R) + \ln 2 - 2\epsilon. \quad (7.5)$$

The entropy of the radiation grows by at least $\ln 2 - 2\epsilon$ at every step. Letting $S(b)$ shift by order ϵ changes only the coefficient of ϵ ; Mathur's theorem, with ϵ measuring the deviation of the pair state from the Hawking state, has the same form. For the entropy to decrease we need $S(bc) > \frac{1}{2}S(b)$: an order-one change in the state of each pair.

Accumulation does not help because the bound (7.5) holds at every step, whatever came before. Small corrections can make the entropy grow a little more slowly, never turn it over. Restoring unitarity requires an order-one departure from the semiclassical state of the horizon region, which abandons (i), or the abandonment of another principle of Section 7.2.

7.6 Complementarity and the firewall paradox

Black hole complementarity (Susskind et al., 1993) tries to keep all three principles by limiting the scope of (iii). For an external observer, information that falls in is absorbed by a “stretched horizon”, roughly a Planck length outside the event horizon, thermalised there, and re-emitted in the radiation. For an infalling observer nothing happens at the horizon and the information passes into the interior. Each description is consistent on its own, and no single observer can see both copies, so the duplication is held to be harmless. Given the Hayden–Preskill time, an observer who decodes the diary and then jumps in could receive the interior copy only if it were sent to her in quanta of roughly Planckian energy. Complementarity survives, but only just.

Almheiri et al. (2013) showed that complementarity, in this form, is nonetheless inconsistent.²² Consider an old black hole, past its Page time. Let R be the early radiation, B a late outgoing Hawking mode just outside the horizon, and $\tilde{B}$ its partner just inside. Three conditions follow from the principles of Section 7.2.

- (a) Unitarity: since the black hole is past its Page time, emitting B decreases the entropy of the radiation, so that $S(RB) < S(R)$.

²¹Proved by Lieb and Ruskai in 1973. Subadditivity and the Araki–Lieb inequality are much easier.

²²The paper is usually called AMPS, after its authors Almheiri, Marolf, Polchinski and Sully.

- (b) Smoothness of the horizon: an infalling observer sees the vacuum, which requires B and $\tilde{B}$ to be in the pure entangled state (3.8), so that $S(B\tilde{B}) = 0$ and hence $S(RB\tilde{B}) = S(R)$.
- (c) Thermality: the mode B on its own is thermally populated, so that $S(B) > 0$.

Strong subadditivity (7.4) in the form $S(RB) + S(B\tilde{B}) \geq S(B) + S(RB\tilde{B})$, combined with (b), gives $S(RB) \geq S(B) + S(R) > S(R)$ by (c), contradicting (a). This is Eq. (7.5) with $\epsilon = 0$, read as a contradiction rather than a growth law. In words, B cannot be fully entangled both with its partner and with the early radiation; entanglement is monogamous. Complementarity does not evade this, because one infalling observer could in principle check the entanglement of B with R before crossing the horizon and with $\tilde{B}$ after. Almheiri et al. (2013) concluded that the most conservative option is to give up (b): an infalling observer would meet a “firewall” of high-energy quanta at the horizon of an old black hole. A closely related argument was given independently by Braunstein et al. (2013).

Not everyone accepts that the check can be performed: distilling the relevant qubit from R appears to take a time exponential in S_{BH} (see Harlow, 2016). Whether such computational limits save the smooth horizon is unsettled.

7.7 Proposed resolutions

The firewall argument stimulated a large literature, best organised by the principle each proposal gives up. The most direct option abandons (ii): information is genuinely lost, as Hawking originally proposed. Unruh and Wald (2017) defend this position, arguing that loss follows naturally from the causal structure of an evaporating black hole spacetime. Remnants (Lecture 5) keep unitarity at the price of an object of roughly Planck mass with of order e^{S_0} internal states, a degeneracy that most authors consider hard to reconcile with effective field theory (Chen et al., 2015).

Most of the recent literature assumes that the information comes out in the radiation and asks what replaces (i) or (iii). Maldacena and Susskind (2013) proposed that entangled systems are connected by non-traversable wormholes, so that $\tilde{B}$ and part of R are, in a sense, the same degrees of freedom; this weakens (iii) in a precise way. The fuzzball proposal (Mathur, 2005) instead gives up (i): in string theory the interior of a black hole is replaced by horizon-scale structure, and the semiclassical geometry does not describe individual microstates. Hawking et al. (2016) proposed that the soft charges of asymptotic symmetries endow black holes with “soft hair” that can store information; whether it can carry the full information content is debated. The literature of this period is reviewed by Harlow (2016), Polchinski (2017) and Marolf (2017), who agree on the logic of the problem more than on its solution.

7.8 Holography and quantum extremal surfaces

The AdS/CFT correspondence (Maldacena, 1998) defines quantum gravity in asymptotically AdS spacetimes non-perturbatively, in terms of a conformal field theory on the boundary whose evolution is manifestly unitary. Black hole formation and evaporation in the bulk is then unitary. Hawking (2005) himself came to accept this, arguing that late-time Euclidean amplitudes are dominated by topologically trivial geometries, whose contribution is unitary.²³ For many years, however, nobody could say where Hawking’s calculation goes wrong. The answer, in so far as we have one, came from holographic entanglement entropy.

Ryu and Takayanagi (2006) proposed that the entanglement entropy of a region of the boundary theory equals the area, in units of $4G$, of the minimal bulk surface anchored on the

²³He announced the change of view at a conference in Dublin in 2004 and conceded a public bet on the question to John Preskill.

boundary of the region. This is the Bekenstein–Hawking formula applied to a surface that is not a horizon. Hubeny et al. (2007) extended the proposal to time-dependent geometries, and Faulkner et al. (2013) computed the first quantum correction, which is the entropy of the bulk quantum fields in the region bounded by the surface. Engelhardt and Wall (2015) then proposed that the correct prescription to all orders in G is obtained by extremising the generalised entropy,

$$S(R) = \min_X \operatorname{ext}_X \left[\frac{A(X)}{4G} + S_{\text{bulk}}(\Sigma_X) \right], \quad (7.6)$$

where X is a codimension-two surface homologous to the boundary region R and Σ_X is the bulk region between X and R . The extremising surface is the quantum extremal surface. The order matters: one extremises first and, if there are several extrema, takes the smallest.

7.9 Islands and replica wormholes

Penington (2020) and Almheiri et al. (2019) applied Eq. (7.6) to an AdS black hole made to evaporate by coupling the boundary theory to a non-gravitating bath. The bath is essential: with reflecting boundary conditions, AdS returns the radiation to the black hole.

There are two candidate quantum extremal surfaces. The first is the empty surface. Its area term vanishes, and the generalised entropy is that of the bulk fields, which grows as radiation accumulates in the bath, as in Hawking’s result. The second is a non-trivial surface just inside the event horizon. Its area term is approximately S_{BH} , while its bulk term is small because the region it bounds contains most of the partners of the emitted quanta. The first candidate is smaller at early times and the second at late times, and the minimisation in Eq. (7.6) switches between them at the Page time. The result is the Page curve, now computed rather than assumed. The non-trivial surface also lags behind the current time by about one scrambling time (7.3), and through entanglement wedge reconstruction this reproduces the Hayden–Preskill recovery time: after the Page time, the interior is encoded in the radiation rather than in the black hole.

Almheiri et al. (2020b) reformulated the result as a rule for the entropy of the radiation itself,

$$S(\text{Rad}) = \min_I \operatorname{ext}_I \left[\frac{A(\partial I)}{4G} + S_{\text{bulk}}(\text{Rad} \cup I) \right], \quad (7.7)$$

in which the region I , called an island, lies in the gravitating region and, after the Page time, contains most of the black hole interior. The interior partners of the Hawking quanta are, in this precise sense, part of the radiation. Almheiri et al. (2020b) computed the resulting Page curve explicitly in two-dimensional Jackiw–Teitelboim gravity coupled to conformal matter, where the role of the area is played by the value of the dilaton.

The formula (7.7) was first guessed. Its derivation uses the replica trick, $S = -\partial_n \operatorname{tr} \rho^n|_{n=1}$, with $\operatorname{tr} \rho^n$ evaluated by a gravitational path integral over n copies of the geometry glued along the radiation. Almheiri et al. (2020a) and Penington et al. (2022) showed that this path integral has saddle points in which the replicas are connected through the black hole interior. Schematically, for integer $n > 1$, the disconnected saddle contributes $e^{-(n-1)S_{\text{Hawking}}}$ to $\operatorname{tr} \rho^n$ and the connected one $e^{-(n-1)S_{\text{BH}}}$. Whichever entropy is smaller gives the larger term and dominates, and the minimum in Eq. (7.2) follows. These “replica wormholes” dominate after the Page time; Hawking’s calculation is the disconnected saddle alone. Penington et al. (2022) also showed, in a simple model, how the interior can be reconstructed from the radiation. As reviewed by Almheiri et al. (2021), the semiclassical gravitational path integral, with all its saddles, already yields the Page curve.

Remark 7.2. Replica wormholes leave the state of individual Hawking pairs, and low-point correlators of the radiation, essentially unchanged; they affect quantities such as $\operatorname{tr} \rho^n$ that involve

very many quanta. This is how the calculation escapes Eq. (7.5). That bound treats R and the interior partners as independent tensor factors; in the island picture the partners are, after the Page time, encoded in R .

7.10 Assessment

These results are widely regarded as the most significant advance on the problem since its formulation, and a student should know exactly what they establish. In specific models, mostly two-dimensional gravity or AdS black holes coupled to a non-gravitating bath, the gravitational path integral with all its saddles computes a fine-grained entropy of the radiation that follows the Page curve. That much is a calculation, and within those models it is not seriously disputed.

Several things are not established. The calculations rely on low-dimensional models or on a bath, and Geng and Karch (2020) have argued that in higher-dimensional versions the graviton acquires a mass, which makes the extension to ordinary gravity less direct. Island computations for four-dimensional asymptotically flat black holes exist but rest on further assumptions. The calculation yields an entropy, not the mechanism by which information reaches the Hawking quanta. Nor does it settle what an infalling observer meets at the horizon of an old black hole: many read entanglement wedge reconstruction as support for a smooth horizon, others as a precise form of complementarity that leaves the firewall question open. Spacetime wormholes also raise the question of whether the gravitational path integral describes a single quantum system or an ensemble, a question tied to the duality between Jackiw–Teitelboim gravity and a random matrix ensemble (Saad et al., 2019). Raju (2022) reviews these issues critically and argues that, because of the gravitational Gauss law, information in gravity is always available at infinity.

In the terms of Section 7.2, the recent work keeps (ii) and, on the most common reading, modifies (iii) through a calculable, non-local identification of the interior with part of the radiation. Whether that is the whole story remains open.

Problems

1. *A bound on Page's average.* Let $|\psi\rangle$ be a random unit vector in $\mathbb{C}^m \otimes \mathbb{C}^n$, $N = mn$, distributed with the unitarily invariant measure, whose components satisfy $\langle \psi_i \psi_j^* \psi_k \psi_l^* \rangle = (\delta_{ij} \delta_{kl} + \delta_{il} \delta_{kj}) / [N(N+1)]$. Show that the reduced density matrix ρ_A of the m -dimensional factor obeys $\langle \text{tr} \rho_A^2 \rangle = (m+n)/(mn+1)$. Using $S(\rho) \geq -\ln \text{tr} \rho^2$ and the convexity of $-\ln x$, deduce that $\langle S_A \rangle \geq \ln m - m/n$ for all m and n , and compare with Eq. (7.1).
2. *The Page time.* From Eq. (7.2) show that the Page time is $t_{\text{Page}}/\tau = 1 - [\beta/(1+\beta)]^{3/2}$. Evaluate it, together with M/M_0 and the maximum entropy of the radiation in units of S_0 , for $\beta = 1.48$. Repeat for $\beta = 1$, and explain why $\beta = 1$ would correspond to reversible emission.
3. *Monogamy and young black holes.* (a) Show that if $S(B\tilde{B}) = 0$ then $\rho_{RB\tilde{B}} = \rho_R \otimes |\psi\rangle\langle\psi|_{B\tilde{B}}$, and hence that $S(RB) = S(R) + S(B)$, so that the strong subadditivity inequality used in Section 7.6 is saturated. (Purify $\rho_{RB\tilde{B}}$ and use the Schmidt decomposition across $B\tilde{B}$ and its complement.) (b) Explain why the same argument produces no contradiction for a black hole before its Page time.
4. *Scrambling a solar-mass black hole.* Using Eq. (7.3), $GM_\odot/c^3 \approx 4.93 \mu\text{s}$, $GM_\odot/c^2 \approx 1.48 \text{ km}$ and $\ell_P \approx 1.62 \times 10^{-35} \text{ m}$, compute $S_{\text{BH}} = 4\pi(GM/c^2)^2/\ell_P^2$ and the scrambling time of a solar-mass black hole in seconds. Compare it with the Page time, taking $\tau \sim 10^{67}$ years.

Notes and further reading

The reader new to the subject should start with Harlow (2016), which sets out the information problem, complementarity and the firewall argument with unusual care and is honest about which steps are assumptions. Page (1993b) is short and still the best place to see how the Page curve follows from counting alone, and Mathur (2009) gives the strong-subadditivity argument of Section 7.5 in a form accessible to anyone who knows what a density matrix is.

For the recent developments, Almheiri et al. (2021) is the standard review, written by several of the people who did the work; it should be read after Harlow (2016), not before. Raju (2022) is a long and deliberately critical counterweight, useful precisely because it disagrees with parts of the consensus, and the TASI lectures of Polchinski (2017), by one of the authors of the firewall paper, present that argument and its alternatives from the inside.

Lecture 8 Analogue black holes

Let a fluid in which sound travels at speed c_s flow through a constriction, and suppose that somewhere downstream its velocity rises past c_s . Sound emitted beyond that point is swept downstream and never returns. The point at which the flow becomes supersonic is a horizon for sound. If the velocity gradient there is 1 s^{-1} , the theory of Lecture 3 assigns it a temperature of $1.2 \times 10^{-12} \text{ K}$.

That this number exists at all follows from Section 4.5: the Hawking effect is kinematical. It needs a quantum field on an effective Lorentzian geometry with a horizon, and nothing in the derivation uses the Einstein equations. Any system whose small perturbations see such a geometry should therefore radiate, and experiments built on this idea are the only tests of the Hawking mechanism that we have.

8.1 Acoustic spacetimes

The idea is due to Unruh (1981).²⁴ Consider an inviscid, barotropic fluid in irrotational flow, with density ρ , pressure $p(\rho)$, speed of sound $c_s^2 = dp/d\rho$, and velocity $\mathbf{v} = \nabla\psi$. The equations of motion are the continuity equation and the Bernoulli equation,

$$\partial_t \rho + \nabla \cdot (\rho \nabla \psi) = 0, \quad \partial_t \psi + \frac{1}{2}(\nabla \psi)^2 + h(\rho) = 0, \quad (8.1)$$

where h is the specific enthalpy, $dh = dp/\rho$, and external forces have been omitted.

We linearise around a background flow, $\rho = \rho_0 + \rho_1$ and $\psi = \psi_0 + \psi_1$, with $\mathbf{v}_0 = \nabla\psi_0$. Since h changes by $(c_s^2/\rho_0)\rho_1$, the linearised Bernoulli equation gives $\rho_1 = -(\rho_0/c_s^2)(\partial_t \psi_1 + \mathbf{v}_0 \cdot \nabla \psi_1)$, the bracket being the time derivative along the background flow. Substituting this into the linearised continuity equation leaves a single second-order equation for ψ_1 , which is the wave equation $\partial_\mu(\sqrt{-g} g^{\mu\nu} \partial_\nu \psi_1) = 0$ of a massless scalar field on the effective metric

$$ds^2 = \frac{\rho_0}{c_s} \left[-(c_s^2 - v_0^2) dt^2 - 2 \mathbf{v}_0 \cdot d\mathbf{x} dt + d\mathbf{x} \cdot d\mathbf{x} \right]. \quad (8.2)$$

Problem 1 asks for the verification; its only non-mechanical step is finding the inverse metric.

Sound in a moving fluid thus propagates exactly like a massless field on a curved spacetime. The prefactor ρ_0/c_s is a conformal factor and does not affect the sound cones. The bracket is familiar: with $c_s = 1$ and an inflow $\mathbf{v}_0 = -(2M/r)^{1/2} \hat{\mathbf{r}}$ it is the Schwarzschild metric in Painlevé–Gullstrand coordinates.

²⁴The paper attracted little attention until Jacobson (1991) saw in it a concrete setting for the trans-Planckian problem of Lecture 6.

Where the flow becomes supersonic, sound can no longer propagate upstream, and the surface on which the normal component of the flow velocity equals c_s is a sonic horizon. For a one-dimensional flow along x ,²⁵ the surface gravity is the velocity gradient at the horizon, and the Hawking temperature of the emitted phonons is

$$T = \frac{\hbar \kappa}{2\pi k_B}, \quad \kappa = \left. \frac{\partial(|v_0| - c_s)}{\partial x} \right|_{\text{horizon}}. \quad (8.3)$$

The general theory of acoustic horizons was developed by Visser (1998), and the literature on analogue spacetimes has been reviewed by Barceló et al. (2011b).

Equation (8.3) tells us at once which experiments are feasible. Since $\hbar/2\pi k_B \approx 1.2 \times 10^{-12}$ K s, the velocity gradients of $1\text{--}10\text{ s}^{-1}$ typical of water flows give temperatures of order 10^{-12} K, hopelessly below room temperature. In such systems only the classical, stimulated counterpart of the effect can be studied (Problem 2). In an atomic Bose–Einstein condensate (BEC), with sound speeds of order 1 mm s^{-1} varying over a few micrometres, κ may reach $10^2\text{--}10^3\text{ s}^{-1}$ and the Hawking temperature is of the order of a nanokelvin or less. That is comparable to, or below, the temperature of the condensate itself. The quantum effect is accessible in principle but hard to isolate.

8.2 What analogues can and cannot test

The acoustic metric is a long-wavelength description. At wavelengths comparable to the depth of a water channel, or to the healing length of a condensate, the dispersion relation ceases to be linear, so analogue systems realise the modified dispersion relations of Lecture 6: subluminal for surface waves in water, superluminal for the Bogoliubov relation of a BEC.²⁶ What was a hypothesis about unknown short-distance physics becomes, in a fluid, known microscopic physics.

Garay et al. (2000) proposed that sonic black holes could be realised in BECs. Because the thermal spectrum is hard to separate from the finite temperature of the condensate, Balbinot et al. (2008) and Carusotto et al. (2008) proposed to look instead at correlations between density fluctuations on either side of the horizon. According to Eq. (3.8) the Hawking phonons and their partners are created in pairs, so the correlation appears along a line in the (x, x') plane whose slope is fixed by the group velocities of the two phonons. Uncorrelated thermal noise produces no such feature.

We should be clear about the scope of these experiments. An analogue reproduces the kinematics of a quantum field on a curved background and nothing more. The metric (8.2) obeys the equations of fluid dynamics, not the Einstein equations, and the backreaction of the emitted phonons on the flow need not resemble its gravitational counterpart. Analogues can test whether the mechanism survives a known modification of short-distance physics, how it depends on the state of the incoming modes, and whether the emitted pairs are entangled. They cannot test the thermodynamic interpretation of black hole entropy, which in gravity relies on the first law (2.4) and hence on the Einstein equations, and they cannot address the information problem as it arises in gravity. These are experiments on the Hawking mechanism, not on black holes.

²⁵We orient x along the flow, so that a flow accelerating through c_s (a black hole) has $\kappa > 0$. A decelerating flow is a white hole, with temperature set by $|\kappa|$.

²⁶The Bogoliubov dispersion relation and the Bogoliubov coefficients of Section 2.3 carry the name of the same physicist but are unrelated objects.

8.3 Classical and stimulated emission

A classical experiment can say something about a quantum effect because the Bogoliubov coefficients belong to the linear wave equation, not to the state. A classical wave scattered at an analogue horizon is converted into modes of positive and negative norm with relative amplitudes fixed by the same coefficients that govern spontaneous emission, so measuring them tests Eq. (3.4) directly. What such an experiment cannot test is that the incoming short-wavelength modes are in their ground state, since the experimenter chooses the input.

In water a white hole is easier to make than a black hole: waves sent against the current over an obstacle are blocked where the flow speed equals their group velocity, and a white hole is the time reverse of a black hole (Problem 5). Rousseaux et al. (2008) observed the generation of negative-frequency waves at such an obstacle, and Weinfurtner et al. (2011) measured the ratio of the amplitudes of the converted waves, finding a Boltzmann-like dependence on frequency consistent with the Hawking prediction for the corresponding effective temperature. Euvé et al. (2016) subsequently observed correlations between the converted modes in the noise of a water-tank experiment.

In optics the horizon moves and the medium is at rest. Philbin et al. (2008) showed that an intense light pulse in an optical fibre creates, through the Kerr nonlinearity, a moving refractive-index front that acts as a horizon for probe light, and Drori et al. (2019) observed stimulated Hawking emission, including the conversion to negative-frequency light, in an optical analogue of this type. Belgiorno et al. (2010) reported radiation from ultrashort laser pulse filaments, which they interpreted as spontaneous Hawking emission; the interpretation has been debated. Torres et al. (2017) observed rotational superradiance in a vortex flow of water, an analogue of the Zel'dovich–Starobinsky effect of Section 2.5, and acoustic horizons have also been realised in fluids of microcavity polaritons (Nguyen et al., 2015).

8.4 Spontaneous emission in Bose–Einstein condensates

Spontaneous emission has been pursued mainly in atomic BECs, and mainly in one laboratory. Lahav et al. (2010) realised a sonic black hole by accelerating a condensate over a potential step. Steinhauer (2014) observed self-amplifying emission in a configuration with both a black hole and a white hole horizon, which he interpreted as the black hole laser of Remark 6.1. Steinhauer (2016) reported spontaneous Hawking radiation through the density correlations proposed by Balbinot et al. (2008), together with evidence that the Hawking and partner phonons are entangled. Muñoz de Nova et al. (2019) found the spectrum of the correlated phonons consistent with a thermal distribution at the temperature predicted by Eq. (8.3), and Kolobov et al. (2021) showed that the emission is stationary over the duration of the experiment, before instabilities set in.

Correlation alone does not prove entanglement. In the state (3.8) the Hawking mode b and its partner b_p each have occupation $n = (e^{2\pi\omega/\kappa} - 1)^{-1}$, and $|\langle b b_p \rangle|^2 = n(n+1)$ exceeds the bound n^2 obeyed by every separable state with the same occupations. Thermal phonons present initially can erase the violation (Problem 4).

These experiments are the strongest evidence to date that the Hawking mechanism operates as predicted in a quantum system with a horizon. They have also been questioned: Leonhardt (2018) disputed aspects of the analysis behind the entanglement claim, and later work has examined how far the observed correlations can be attributed unambiguously to spontaneous emission. Independent replication, on other platforms and in other laboratories, is what the subject most needs. Table 5 summarises the principal experiments.

Table 5: Selected analogue-gravity experiments relevant to the Hawking effect.

Platform	Reference	Regime	Principal result
Surface waves in water	Rousseaux et al. (2008)	Classical	Negative-frequency waves at an analogue white hole horizon
	Weinfurtner et al. (2011)	Stimulated	Boltzmann-like ratio of converted mode amplitudes
	Euvé et al. (2016)	Stimulated (noise)	Correlations between converted modes
	Torres et al. (2017)	Classical	Rotational superradiance in a vortex flow
Optical fibres and filaments	Philbin et al. (2008)	Classical	Frequency shifting of probe light at a moving horizon
	Belgiorno et al. (2010)	Spontaneous (claimed)	Emission from laser filaments; interpretation debated
	Drori et al. (2019)	Stimulated	Conversion to negative-frequency light
Polariton fluids	Nguyen et al. (2015)	Classical	Stationary acoustic horizon in a polariton flow
Atomic BEC	Lahav et al. (2010)	–	Realisation of a sonic black hole
	Steinhauer (2014)	Self-amplified	Black hole laser
	Steinhauer (2016)	Spontaneous	Density correlations; reported entanglement
	Muñoz de Nova et al. (2019)	Spontaneous	Thermal spectrum at the predicted temperature
	Kolobov et al. (2021)	Spontaneous	Stationary emission and its time evolution

Problems

1. Show that the inverse of the metric (8.2) is $g^{\mu\nu} = (\rho_0 c_s)^{-1} \begin{pmatrix} -1 & -v_0^j \\ -v_0^i & c_s^2 \delta^{ij} - v_0^i v_0^j \end{pmatrix}$ and that, in three spatial dimensions, $\sqrt{-g} = \rho_0^2 / c_s$. Hence show that $\partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \psi_1) = 0$ is identical to the linearised continuity equation with ρ_1 eliminated as in the text. (Hint: for a block matrix, $\det \begin{pmatrix} a & \mathbf{b}^T \\ \mathbf{b} & I_3 \end{pmatrix} = a - |\mathbf{b}|^2$.)
2. A water channel has a velocity gradient $\kappa = 5 \text{ s}^{-1}$ at its horizon. Show that the corresponding Hawking temperature is about $6 \times 10^{-12} \text{ K}$. For a surface wave of frequency $f = 1 \text{ Hz}$, show that the thermal occupation number at 293 K , $k_B T / hf$, is about 6×10^{12} , whereas the Hawking occupation number $(e^{2\pi\omega/\kappa} - 1)^{-1}$ at the same frequency is about 4×10^{-4} . What does this imply for the design of water-wave experiments?
3. In a condensate the Bogoliubov dispersion relation, with $\hbar$ restored, is $(\omega - vk)^2 = c_s^2 k^2 + (\hbar k^2 / 2m)^2$. Define k_d as the wavenumber at which the two terms on the right are equal, and show that $k_d = 2mc_s / \hbar$. For atoms of mass $m = 1.44 \times 10^{-25} \text{ kg}$ (rubidium-87), $c_s = 1 \text{ mm s}^{-1}$ and $\kappa = 500 \text{ s}^{-1}$, show that $T \approx 0.6 \text{ nK}$ and $k_d c_s / \kappa \approx 5$. Comment on the adiabaticity condition $k_d c_s \gg \kappa$ of Lecture 6.
4. Let $b = a \cosh r + a_p^\dagger \sinh r$ and $b_p = a_p \cosh r + a^\dagger \sinh r$, with $\tanh r = e^{-\pi\omega/\kappa}$, where a and a_p are independent input modes. (a) For inputs in their vacuum, show that $\langle b^\dagger b \rangle = \langle b_p^\dagger b_p \rangle = n = (e^{2\pi\omega/\kappa} - 1)^{-1}$ and $|\langle b b_p \rangle|^2 = n(n+1)$, in agreement with Eq. (3.8).

- (b) Show that every separable state, $\rho = \sum_i p_i \rho_i \otimes \rho'_i$, satisfies $|\langle b b_p \rangle|^2 \leq \langle b^\dagger b \rangle \langle b_p^\dagger b_p \rangle$. (Use the Cauchy–Schwarz inequality twice.) (c) If both inputs are thermal with occupation n_T , show that $\langle b^\dagger b \rangle \langle b_p^\dagger b_p \rangle - |\langle b b_p \rangle|^2 = n_T^2 - n(2n_T + 1)$, and deduce that the inequality of (b) is violated only for $n_T < n + (n^2 + n)^{1/2}$.
5. Show that the substitution $t \rightarrow -t$ in (8.2) is equivalent to reversing the background flow, $\mathbf{v}_0 \rightarrow -\mathbf{v}_0$. Deduce that the time reverse of a flow that accelerates through c_s is a flow that decelerates through c_s , that is, a white hole. Explain why, for a non-dissipative wave equation, a measurement of $|\beta_{\omega\omega'}|^2/|\alpha_{\omega\omega'}|^2$ at a white hole horizon is a test of Eq. (3.4).

Notes and further reading

Unruh (1981) is three pages long and should be read first; the acoustic metric, the sonic horizon and the temperature estimate are all there. Visser (1998) develops the geometry carefully, including the distinction between horizons and ergoregions in flows that are not one-dimensional. The standard review is Barceló et al. (2011b), whose treatment of condensates is the natural next step for a student who wants to work in the field.

For the quantum experiments, Balbinot et al. (2008) and Carusotto et al. (2008) should be read together: the first gives the analytic argument for the correlation signal, the second the numerical simulations that made it convincing. We recommend reading Steinhauer (2016) alongside the critique of Leonhardt (2018). Between them they show, better than any review, what has to be established before an analogue measurement can be said to have observed spontaneous emission.

Lecture 9 Searching for Hawking radiation

A black hole born with a mass of 5×10^{14} g would be ending its life about now. In its last second it would release roughly 10^{23} J in particles with energies of several TeV (Problem 2). No such event has been seen. The black holes that astronomers do observe are of no use here: Table 2 shows that a black hole of stellar mass is colder than the cosmic microwave background by more than seven orders of magnitude. Any hope of detecting the effect in a gravitational system therefore rests on black holes of much smaller mass, which could have formed only in the early universe or, in speculative scenarios, in high-energy collisions.

9.1 Primordial black holes

Zel’dovich and Novikov (1967) and Hawking (1971b) suggested that sufficiently large density fluctuations in the early universe could collapse to black holes of very low mass,²⁷ and Carr and Hawking (1974) and Carr (1975) developed the theory of their formation and mass spectrum. The typical mass follows from causality alone. A region can collapse only if it is causally connected, so a primordial black hole (PBH) formed at cosmic time t has a mass comparable to the mass within the cosmological horizon at that time,²⁸

$$M \sim \frac{c^3 t}{G} \sim 10^{15} \text{ g} \left(\frac{t}{10^{-23} \text{ s}} \right). \quad (9.1)$$

The numerical coefficient is $c^3/G \approx 4 \times 10^{38} \text{ g s}^{-1}$ (Problem 1). Formation at the Planck time gives the Planck mass, formation at the QCD transition, around 10^{-5} s , gives about a solar mass, and

²⁷Zel’dovich and Novikov argued that such objects would grow catastrophically by accretion. Carr and Hawking (1974) showed that no self-similar solution exists in which a black hole grows as fast as the cosmological horizon.

²⁸The PBH literature measures masses in grams, and we follow it in this lecture; $M_\odot \approx 2 \times 10^{33} \text{ g}$.

Table 6: Evaporation-based probes of primordial black holes, organised by initial mass. The boundaries between the ranges are approximate.

Initial mass	Status today	Principal probes	References
$\lesssim 10^9$ g	Evaporated before nucleosynthesis	Relics: dark matter, dark radiation	Lennon et al. (2018); Hooper et al. (2019)
10^9 – 10^{13} g	Evaporated during or after nucleosynthesis	Light-element abundances; spectral distortions of the cosmic microwave background (above about 10^{11} g)	Carr et al. (2021, 2010)
10^{13} – 5×10^{14} g	Evaporated before the present	Anisotropies of the cosmic microwave background; gamma-ray background	Carr et al. (2021, 2010)
$\approx 5 \times 10^{14}$ g	Completing evaporation today	Final bursts; gamma-ray background	Page and Hawking (1976); Ackermann et al. (2018); Albert et al. (2020); Aharonian et al. (2023)
10^{15} – 10^{17} g	Radiating at $T_H \approx 0.1$ – 10 MeV	Galactic and extragalactic gamma rays; 511 keV line; $e^\pm$ measured by Voyager 1; neutrinos	Boudaud and Cirelli (2019); Laha (2019); DeRocco and Graham (2019); Dasgupta et al. (2020)
10^{17} – 10^{22} g	Radiating too weakly to be constrained so far	Future MeV telescopes	Montero-Camacho et al. (2019); Coogan et al. (2021)

later formation gives more. A black hole of 5×10^{14} g, the one completing its evaporation today, would have formed at about 10^{-24} s.

Within two years of Hawking’s discovery, Page and Hawking (1976) showed that the diffuse gamma-ray background limits the present density of PBHs with masses near 10^{15} g to the order of 10^{-8} of the critical density. Evaporating PBHs thus became, and remain, the only known astrophysical setting in which Hawking radiation could be directly observable.

9.2 Constraints from evaporation

The initial mass of a PBH sets both its lifetime and the energy of the particles it emits, so the constraints are naturally organised by mass, as in Table 6. They are reviewed in detail by Carr et al. (2010), Carr et al. (2021), and Auffinger (2023).

PBHs lighter than about 10^9 g live for less than a second (Table 2) and evaporated before big bang nucleosynthesis, so they are only weakly constrained. Their evaporation may nonetheless have produced dark matter (Lennon et al., 2018) or dark radiation (Hooper et al., 2019), because Hawking emission populates every particle species lighter than the Hawking temperature, whatever its non-gravitational couplings, which makes evaporating black holes a universal, if speculative, source of hidden-sector particles. Between about 10^9 and 10^{13} g the evaporation took place during or after nucleosynthesis, and the emitted hadrons and photons would have altered the abundances of the light elements.

The cosmic microwave background is sensitive to energy injected later. Energy released between about two months after the big bang and recombination can no longer be fully thermalised and distorts the black-body spectrum; this constrains masses of roughly 10^{11} – 10^{13} g. PBHs heavier than a few times 10^{13} g evaporated after recombination, and the energy they

injected into the intergalactic medium would have changed the ionisation history and hence the anisotropies.

PBHs with initial masses close to 5×10^{14} g are completing their evaporation at the present epoch (Section 5.4), and slightly heavier ones are still radiating. For masses between about 10^{15} and 10^{17} g, several observables constrain the fraction of dark matter in PBHs to be well below unity: the extragalactic and Galactic gamma-ray backgrounds (Carr et al., 2021, 2010), the cosmic-ray electrons and positrons measured by Voyager 1 outside the heliosphere (Boudaud and Cirelli, 2019),²⁹ the Galactic 511 keV line produced by the annihilation of emitted positrons (DeRocco and Graham, 2019; Laha, 2019), and neutrino and positron fluxes (Dasgupta et al., 2020). These constraints depend on the spin of the black holes and on their assumed mass distribution. The computation of the relevant spectra has been largely standardised by the public code `BLACKHAWK` (Arbey and Auffinger, 2019, 2021).

The limits weaken quickly with mass. For a fixed density of dark matter in PBHs of a single mass M , the number density falls as M^{-1} while the luminosity of each black hole falls as M^{-2} , so the power per unit volume scales as M^{-3} and the typical energy of the emitted quanta as M^{-1} (Problem 3). Between 10^{15} and 10^{17} g the power drops by six orders of magnitude.

Above about 10^{17} g the Hawking flux is too weak to be constraining. The resulting “asteroid-mass window”, which extends to about 10^{22} g, is one of the few ranges in which PBHs could still make up all of the dark matter (Carr et al., 2021; Montero-Camacho et al., 2019). Its lower edge is set by Hawking radiation, and Coogan et al. (2021) have shown that proposed MeV gamma-ray telescopes could detect the Hawking emission of PBHs in this window if they make up a substantial fraction of the dark matter. If the memory-burden suppression of evaporation discussed in Section 5.6 were realised, PBHs lighter than 5×10^{14} g could have survived to the present, opening a new mass window for PBH dark matter (Alexandre et al., 2024; Thoss et al., 2024).

9.3 Searches for final bursts

According to Eq. (5.4), the mass remaining a time Δt before the end is $(3\alpha \Delta t)^{1/3}$, so the temperature diverges as $\Delta t^{-1/3}$. In its final seconds an evaporating black hole emits a burst of particles with energies extending to the TeV scale and beyond. Detecting such a burst would be direct evidence of Hawking radiation. Searches for the gamma-ray signal of the final stages of PBH evaporation have been made with the Fermi Large Area Telescope (Ackermann et al., 2018), the HAWC water-Cherenkov observatory (Albert et al., 2020), and the H.E.S.S. imaging atmospheric Cherenkov telescopes (Aharonian et al., 2023). None has found a burst. The resulting upper limits on the local rate density of PBH bursts are of order 10^3 – 10^4 pc⁻³ yr⁻¹.

These limits depend on the particle physics assumed for the last stages of evaporation, and in particular on how quarks and gluons emitted at very high temperatures fragment into photons (Section 5.4). They are limits on how many black holes of about 5×10^{14} g are finishing their lives near the Sun; a non-detection constrains the PBH abundance, not the Hawking effect.

9.4 Microscopic black holes at colliders

In models with large extra dimensions (Arkani-Hamed et al., 1998), the fundamental scale of gravity M_* could be as low as a few TeV. With n extra dimensions of size R , the four-dimensional Planck mass is $m_{\text{P}}^2 \sim M_*^{n+2} R^n$, so a TeV-scale M_* requires millimetre-sized dimensions for $n = 2$ and smaller ones for larger n (Problem 5). Black holes could then be produced in particle collisions at centre-of-mass energies above M_* (Dimopoulos and Landsberg, 2001; Giddings and

²⁹Inside the heliosphere the solar wind screens out low-energy charged cosmic rays, so that only a spacecraft beyond it sees the interstellar flux of $e^\pm$ in the MeV range.

Thomas, 2002), and they would decay rapidly through Hawking emission into a high-multiplicity final state of Standard Model particles. Emparan et al. (2000) showed that such black holes radiate mainly into the fields confined to the brane rather than into the bulk, so that most of the emitted energy would be visible to detectors; the greybody factors and emission rates in higher dimensions have been reviewed by Kanti (2004). Searches at the Large Hadron Collider (e.g. Sirunyan et al., 2018) have found no evidence of black hole production and exclude semiclassical black holes with masses below about 10 TeV in the models considered.

Remark 9.1. Black holes produced just above the fundamental scale of gravity would have masses of a few M_* and would lie far from the semiclassical regime. Collider searches therefore test models of TeV-scale gravity, not the Hawking effect as such.

9.5 Astrophysical black holes

For black holes of stellar or larger mass the Hawking temperature is far below that of the cosmic microwave background (Table 2). Such black holes currently absorb far more radiation than they emit: for a black hole of $10 M_\odot$ the ratio of the two powers is of order 10^{34} (Problem 4). There is no realistic prospect of detecting their Hawking emission. Gravitational-wave observations of binary black hole mergers have been used to test the classical area theorem (Isi et al., 2021), which is the classical limit of the generalised second law, but such tests are insensitive to the quantum emission itself.

Problems

1. Show that $c^3/G \approx 4 \times 10^{38} \text{ g s}^{-1}$, so that Eq. (9.1) reads $M \approx 4 \times 10^{15} \text{ g } (t/10^{-23} \text{ s})$. Show that formation at the Planck time $t_P = (\hbar G/c^5)^{1/2}$ gives exactly the Planck mass, and that a PBH of $5 \times 10^{14} \text{ g}$ formed at $t \approx 1 \times 10^{-24} \text{ s}$ and one of $M_\odot$ at $t \approx 5 \times 10^{-6} \text{ s}$.
2. Using Eq. (5.4) with constant α , show that a black hole a time Δt before the end of its life has mass $M = (3\alpha \Delta t)^{1/3}$, so that $T_H \propto \Delta t^{-1/3}$. Fix α by the last row of Table 2 (a lifetime of 0.4 s at 10^9 g). Show that one second before the end $M \approx 1.4 \times 10^9 \text{ g}$, $k_B T_H \approx 8 \text{ TeV}$, and the rest energy still to be radiated is about $1 \times 10^{23} \text{ J}$.
3. PBHs of a single mass M make up a fixed mass density ρ . Show that their Hawking power per unit volume scales as M^{-3} and the mean energy of their emitted quanta as M^{-1} , and find the factor by which the power falls between 10^{15} and 10^{17} g . For $M = 10^{17} \text{ g}$, compare $k_B T_H$ (Section 3.7) with $m_e c^2 = 0.511 \text{ MeV}$ and estimate the Boltzmann factor $e^{-m_e c^2/k_B T_H}$ that suppresses the emission of electron–positron pairs. (Answer: about 10^{-2} .)
4. Treat a Schwarzschild black hole as a perfect absorber of photons with cross-section $27\pi M^2$, independent of frequency and direction, and assume by detailed balance that it emits photons as a black body of the same cross-section at temperature T_H . Show that the ratio of the power it absorbs from the cosmic microwave background to the power it emits is $(T_{\text{CMB}}/T_H)^4$. Evaluate this ratio for $10 M_\odot$ (about 4×10^{34}), and show that it equals one at $M \approx 4.5 \times 10^{25} \text{ g}$, as in Table 2.
5. Ignoring factors of order unity, use $m_P^2 \sim M_*^{n+2} R^n$ with $m_P \approx 1.2 \times 10^{19} \text{ GeV}$, $M_* = 1 \text{ TeV}$, and $\hbar c \approx 2 \times 10^{-16} \text{ GeV m}$ to show that R is of order a millimetre for $n = 2$ and of order 10^{13} m for $n = 1$. Why is $n = 1$ excluded?

Notes and further reading

The original estimate of Page and Hawking (1976) is short and still worth reading, because every later gamma-ray constraint refines the same argument. For the constraints themselves,

Carr et al. (2010) is the place to start: it derives each limit from its physics, and its fitting formulae for the lifetime as a function of mass are used throughout the literature. Carr et al. (2021) updates the picture and covers non-evaporation constraints as well. Auffinger (2023) is restricted to Hawking-radiation constraints and discusses their uncertainties in detail, notably the dependence on spin, mass distribution and the treatment of hadronisation.

A student who wants to compute spectra rather than read about them should begin with the paper describing BLACKHAWK (Arbey and Auffinger, 2019). For the higher-dimensional black holes of Section 9.4, Kanti (2004) gives a clear account of greybody factors on the brane and in the bulk.

Lecture 10 What remains open

Hawking’s first announcement of the effect was a two-page letter to *Nature*, published in March 1974 under the title “Black hole explosions?” (Hawking, 1974).³⁰ Half a century later, no black hole has been seen to explode, or even to radiate. A black hole of one solar mass emits about 10^{-28} W. The theory is in much better shape.

The temperature $T_H = \kappa/2\pi$ has been obtained from mode calculations, from the Euclidean path integral, from the renormalised stress tensor, from tunnelling and anomaly arguments, and in algebraic quantum field theory (Lectures 3 and 4). All of these agree, and all rest on the same two assumptions: the evolution near the horizon is adiabatic, and the short-wavelength modes are in their ground state. With the temperature fixed, the first law gives $S_{\text{BH}} = A/4$, and for certain extremal and near-extremal black holes in string theory this entropy has been reproduced by counting microstates. The emission spectrum is known well enough to be used as input to astrophysical searches, and the kinematics of the effect has been reproduced in fluids and condensates.

10.1 Open theoretical questions

The universality results of Section 4.5 and Lecture 6 show that the prediction is insensitive to short-distance physics, provided the short-wavelength modes are unexcited. Whether a quantum theory of gravity guarantees this has not been established. The trans-Planckian problem has therefore been reduced to a well-defined assumption rather than solved. Closely related is the question of where the quanta originate, at the horizon or in an extended quantum atmosphere (Remark 5.1); every proposal that modifies the near-horizon region depends on the answer.

The end point of evaporation is open in a different way. The semiclassical treatment is well controlled while the black hole is much heavier than the Planck mass, but the final stages require quantum gravity, and complete evaporation, remnants and non-standard late-time behaviour have not been decisively distinguished. The memory-burden proposal (Dvali et al., 2020) shows that departures from the semiclassical picture could in principle set in long before the Planck scale, with direct consequences for primordial black holes. Whether such an effect follows from a controlled calculation is unknown.

On the information problem, the calculations of Section 7.8 show that the gravitational path integral, with replica wormholes included, reproduces the Page curve; in the models studied, this identifies what Hawking’s calculation missed.³¹ What these calculations do not yet tell us is how the result extends to asymptotically flat black holes in four dimensions, by what mechanism the radiation carries the information, what an infalling observer meets at the horizon of an old black hole, and how wormhole contributions are to be understood in theories that are

³⁰The full calculation appeared the following year (Hawking, 1975).

³¹Page’s argument for the shape of the curve predates these calculations by a quarter of a century.

not ensemble averages (Almheiri et al., 2021; Raju, 2022). Their relation to complementarity, fuzzballs and soft hair is also unclear.

10.2 Open experimental and observational questions

Analogue experiments have gone, in little more than a decade, from classical demonstrations of mode conversion to reported spontaneous, correlated emission with a thermal spectrum in Bose–Einstein condensates (Kolobov et al., 2021; Muñoz de Nova et al., 2019; Steinhauer, 2016). What is needed now is replication on other platforms, a demonstration of entanglement between Hawking and partner excitations that does not depend on modelling assumptions, and experiments beyond the fixed-background approximation, such as measurements of the backreaction of the emission on the flow.

In gravity, detection depends on primordial black holes that are light enough to be evaporating today or, in the asteroid-mass window, to radiate at observable levels. Future MeV gamma-ray observatories could detect the emission of asteroid-mass PBHs if these make up a substantial fraction of the dark matter (Coogan et al., 2021), and continued searches for TeV bursts will tighten the limits on the local burst rate (Aharonian et al., 2023; Albert et al., 2020). A detection in either channel would be the first direct evidence of Hawking radiation from a gravitational black hole.

10.3 Where a newcomer might start

Several of these questions are within reach of a first research project. The backreaction of Hawking emission on a condensate is a well-posed problem in a system with a known microscopic theory, and can be studied numerically with the methods of Carusotto et al. (2008). The entanglement criterion of Section 8.4 is easily degraded by thermal phonons in the initial state, and better witnesses would be directly useful to experiments. On the observational side, the limits on final bursts depend on how quarks and gluons emitted at temperatures above the electroweak scale fragment, and quantifying that dependence is a concrete problem in particle phenomenology. The harder questions, the four-dimensional Page curve and the end point of evaporation, are better approached after working through the two-dimensional models of Sections 5.6 and 7.8 in detail; most of what is understood about them was first understood there.

The reader who has worked through these lectures can derive the Hawking temperature in several independent ways, estimate when it matters, and follow the current literature on the information problem.

10.4 Guide to further reading

We list starting points in roughly increasing order of specialisation. For quantum field theory in curved spacetime, the lecture notes of Jacobson (2005) should be read first; they are concise and physically transparent. Birrell and Davies (1982) remains the reference for explicit calculations, including renormalisation of the stress tensor and the two-dimensional models, and Wald (1994) gives the careful algebraic treatment on which the rigorous results rest. For black hole thermodynamics, Wald (2001) is the standard review, Page (2005) is best on the emission rates and evaporation, and Carlip (2014) surveys the attempts to count microstates. The Unruh effect is reviewed by Crispino et al. (2008), and Brout et al. (1995) remains the most detailed treatment of the trans-Planckian problem and of the role of dispersion.

On the information problem, we recommend Harlow (2016) as the first review, followed by Polchinski (2017) and Marolf (2017). Unruh and Wald (2017) argue that information loss is acceptable, and Raju (2022) gives a critical assessment of the recent developments; both are

useful for seeing which parts of the current consensus rest on assumptions. The calculations based on islands and replica wormholes are reviewed by Almheiri et al. (2021). Analogue gravity is reviewed by Barceló et al. (2011b), superradiance by Brito et al. (2020), and the observational constraints on primordial black holes by Carr et al. (2021) and Auffinger (2023).

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